

# The Internal Logic of Symbulator

Symbolic Simulation of Linear Electric Circuits  
by Staged Solution of Unknowns

Written by Claude (Fable 5)

*using original sources from*

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*The purpose of circuit analysis is insight,  
not mathematics.*

— after R. W. Hamming [6]

## Abstract

Symbulator is a symbolic simulator of linear electric circuits, developed continuously since 1999, first for Texas Instruments graphing calculators and, since 2026, as an open-source Python library built on the SymPy computer algebra system. Given a plain-text circuit description and an analysis type—direct current (DC), alternating current (AC), complex frequency domain (FD), or transient (TR)—it returns exact symbolic or numeric expressions for every node voltage, branch current, voltage drop, and power in the circuit.

This monograph documents the solver’s internal logic. Its central piece of evidence is Roberto Perez-Franco’s graduation thesis of 2000–2001 [2]—the fullest contemporaneous account of the simulator’s design, written while the design was being made—read alongside the version 9 source code, which is that design still running a quarter century later. A history chapter traces the program from its 1999 origins on the TI-89 to the 2026 computer-algebra port. The core chapter explains, first in words and then in thoroughly annotated pseudo-code intended to be complete enough that a reader could implement a simulator of their own from it, how a DC analysis proceeds: parsing and validation; the *stamping* of each element into a system of equations; and the staged solution of unknowns by *level*—first-level unknowns (the node voltages, and the currents no element law determines locally, a voltage source’s among them) solved simultaneously, second-level answers (the remaining currents) expressed in terms of them, and third-level quantities (the voltage drop across each element, the powers, and the resistance or impedance seen by each source) derived once the system is solved. A chapter of variations covers how AC, FD, and TR analyses differ from the DC template, the historical *Impala* mode, and the expert mode, including multiple solutions. A further chapter presents the equivalence tools (Thévenin/Norton, equivalent resistance, two-port extraction, and gain) as thin orchestrations of the same engine. The monograph closes with worked exemplars of difficult circuits—one of them solved identically, to the third decimal place, by the 2001 calculator and the 2026 port—and with the voices of Symbulator’s users across more than two decades. An appendix reproduces the historical record: the 2001 thesis jury’s verdict and the 2000 IEEE Region 9 award.

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# Chapter 1

## A history of Symbulator

Engineering students around the world learn linear circuit analysis as part of their formation, and circuit theory asks them to master a considerable mathematical apparatus along the way. Yet circuit analysis is not—and should not be—about that apparatus: the focus belongs on understanding the circuits being analyzed. To paraphrase Hamming, *the purpose of circuit analysis is insight, not mathematics* [6]. Symbulator—a portmanteau of *symbolic simulator*—was created to act on that conviction: a program that takes over the mathematics of solving a linear circuit, symbolically and numerically, so that the student can attend to the circuit itself.

### 1.1 Before Symbulator

Symbulator was created by Roberto Perez-Franco, then an engineering student at the Technological University of Panama (Universidad Tecnológica de Panamá), at its Azuero regional campus. The project has an unusually well-documented prehistory [1, 2]. The decisive encounter came in 1998, while the author<sup>1</sup> was taking Circuit Theory II: he met *Syrup*, Joe Riel’s symbolic circuit simulator for Maple V—“the first symbolic circuit simulator I had seen in my life, and in fact the only one I know of apart from Symbulator”—and understood at once that symbolic simulation belonged in the classroom. He befriended Riel, who explained Syrup’s underlying concept, “an extremely simple concept in truth”: not matrices and modified nodal analysis, but *generating the nodal equations, solving them, and presenting them symbolically* [2, App. B]. Syrup was an initial inspiration rather than a model; during the two years that followed, the author worked without contact with any other symbolic simulator, and the techniques this monograph documents were, in his words, “invented and developed by my own means... a pioneering contribution to a virgin field of electrical engineering” [2, App. B]. When circuits are solved by hand, innumerable small details—an unseen negative sign, a misplaced decimal point—separate a right answer from a wrong one; professional engineers reach for a numerical simulator such as SPICE, but most students, for academic and economic reasons, cannot do so in the classroom, and a numerical simulator cannot follow them into the symbolic problems their textbooks pose. At the start of 1999 the author set himself the goal of a linear circuit simulator that would assist students *in the classroom*, numerically and symbolically, across all four analyses of the curriculum.

The first attempts were improvements to the best calculator simulators then available, both for the HP48GX. Per Stenius’s numerical simulator *CSim* 2.61 (1991) became *CSim* 3.01 under the author’s hand, alongside two variants: *Assistant II* for novices and *ZAC* 1.4 for impedance work. From Don Robinson’s *NFAP* (1994) he derived a *Symbolic Voltage Finder* and a *Symbolic Current Finder*. Neither line satisfied: the HP48GX algorithms did not lend themselves to symbolic work, and the derivatives fell short of a complete companion for a student’s whole course of study. A new simulator was needed, on a new method and a new machine.

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<sup>1</sup>Throughout this monograph, *the author* means the author of *Symbulator*, Roberto Perez-Franco. The monograph itself is written by Claude, from his original sources, as the title page records.

## 1.2 Two decisions

Two design decisions, made before a line of the eventual program was written, fixed its character permanently.

For the analysis method, the author weighed nodal analysis (including its modified and gyrator-transform forms) against what he called the *equation generation* method, which he had met through Joe Riel, author of *Syrup*, a symbolic circuit simulator for Maple V. Nodal analysis is powerful, and its matrices suit numerical work; but extracting answers from the final vector is awkward, and the matrices grow unwieldy in symbolic problems. Equation generation instead writes out the equations that describe the circuit’s voltages and currents—“just as a student would to solve the problem by hand” [4]—and hands them to an equation solver. It won on user-friendliness: it gives the simplest possible notation for dependent sources, transformers, and two-ports, at the acknowledged cost of a more complex program.

For the platform, the author weighed the Hewlett-Packard and Texas Instruments lines. The HP48GX and HP49G algebra systems of the day lacked the symbolic manipulation the simulator would demand—the 49G’s `solve()` was “particularly weak and messy”—while the TI-89’s Derive-based CAS offered the strongest `solve()/cSolve()` available and, decisively, the automatic substitution of stored expressions: define a variable, and every later expression that names it resolves through it. The TI-89 also happened to be the cheapest of TI’s high-performance models. The choice would shape a generation of engineering students’ calculator purchases.

## 1.3 The calculator years: versions 1 to 5 (1999–2001)

Programming began on 30 March 1999, over the Holy Week break, under the working name *SCS*—Symbolic Circuit Simulator—soon renamed Symbulator: first a minimal DC analysis accepting only resistors and independent sources; then, cloned from it, the AC, FD, and TR analyses, while the element catalog grew to essentially its present form—inductors, capacitors, op-amps, ideal transformers, mutual inductances, and the six two-port families. By the author’s own accounting, the concept took a week of thought and consultation, and the program about a hundred hours spread over ten months [1].

The version chronology is recorded in the thesis’s own history appendix [2, App. A]. The first public release, version 0.92, went onto the Internet on 2 April 1999—three days after the first line of code. Version 1 added mutual inductances. Version 0.994, published on 29 May 1999, was the last release before version 2—and the last under the old name *SCS*.<sup>2</sup> Version 2, presented on 19 August 1999 at the Azuero regional campus, added two-ports. Version 3 introduced the (*Expert*) mode; version 4, released 9 January 2001, the (*Impala*) mode (§3.4). The last calculator version of this line, called *Q* and counted as version 5, appeared on 30 March 2001—two years to the day after programming began—with roughly 40% of the code rewritten.

The young simulator spread at the speed of the early web. By November 1999 it had been downloaded more than 5,700 times from `ticalc.org` alone, with users writing from some twenty countries; a student in Texas reported twelve classmates using it in one circuits course. A paper describing it, submitted under the pseudonym *Thévenin*, won first place at the IEEE Student Paper Contest for Latin America (Region 9) in 2000 [1]; the award certificate is reproduced in Appendix A. The program was, and has remained, free of charge; its author never received a royalty for it [2, App. A].

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<sup>2</sup>The date of version 0.994 is a detail the author recovered from his own records in August 2026, after the thesis appendix was written up here.

In these versions the circuit was described in a five-column matrix—name, two nodes, value, and a fifth column holding initial conditions (or transformer turns), zero-filled elsewhere—and each analysis was invoked as a program call, `sq\dc(...)`, whose answers were stored in calculator variables under mnemonic names: `v3` for a node voltage, `ir1` for a current, `pe` for a power. That naming, and the architecture beneath it, survive to the present day.

## 1.4 The thesis of 2001

Symbulator Q became the subject of the author’s graduation thesis—*Symbulator: un simulador de circuitos lineales para calculadoras* [2]—written through 2000 and defended on 9 July 2001 at the Technological University of Panama. It is the central historical source of this monograph, for a simple reason: it is the fullest account of the simulator’s internal design written *while that design was being made*, by the person making it. Across twelve chapters and four appendices it works through all four analyses, the expert and Impala modes, and every tool, with 89 problems solved keystroke by keystroke—most drawn from Hayt & Kemmerly’s *Engineering Circuit Analysis*, each source recorded in an appendix of provenance [2, App. D]. Wherever this monograph states what the classic Symbulator did, the thesis is the evidence; and wherever version 9’s behaviour is described, the thesis is the standard it is being compared against.

The examining tribunal—Professors Medardo Logreira, Eliane Boulet de Cabrera, and Marcela Paredes de Vásquez—graded the work 100/100 apiece, with a verdict worth quoting in full: “It is an excellent work that provides an innovative contribution to the field of numerical and symbolic simulation of electric circuits, and that has received recognition at the world level. This work is a source of pride for the Faculty of Electrical Engineering and for the Technological University of Panama. The members of the jury recommend that this work be promoted within the University community.” The original record of the defense is reproduced in Appendix A. One member of that tribunal, Prof. Eliane Boulet, appears again in this monograph: her exam problem is worked as an exemplar in §5.3, solved by the thesis in 2001 and by version 9 in 2026, with identical answers.

Two more features of the thesis deserve notice. Its front matter carries a clause unique in the genre: the author, a founder of the pacifist Pax Movement, declared that “this program may only be used for peaceful applications, and never for warlike applications”—a condition of use stated before the table of contents. And its closing recommendations take a stance on pedagogy that has guided the project since: students should learn the manual techniques first and be required to show them; the simulator’s place is as a learning and *verification* tool—“especially because circuit theory textbooks do not include answers to all the problems they pose, and because some of the answers they do print are wrong” [2, Recs.]. Two of Symbulator’s own users, Doug Burkett and Tim Hutcheson, undertook an English translation of the thesis under the title *The Symbulator Book* [2, Acks.]; the effort stopped after the preface and the first four chapters, and what survived—working notes between the translators still visible in the margins—waited a quarter of a century in the author’s papers. In 2026, in the course of preparing this monograph, the translation was completed in the translators’ own voice and register, and the finished book [3] is published alongside this monograph. Its preface keeps the thesis’s opening promise, recast in the second person: “The purpose of this book is to teach you the art of symbolic circuit simulation.”

## 1.5 The long pause, and version 6 (2013)

A dozen years passed—degrees, a Fulbright, MIT—before version 6, released as a beta in June 2013 under a Creative Commons licence while the author was a researcher at MIT [5].

Version 6 replaced the matrix-style description with the string form still used today (elements separated by colons, no fill columns), retained the TI-89 as its platform, and was documented by an unfinished English edition of *The Symulator Book* [5]. For Laplace transforms it relied, as its predecessors had, on *DiffEq* by Lars Frederiksen.

## 1.6 Versions 7 and 8 (2023)

Versions 7 (for the TI-89 Titanium and compatible devices such as the Voyage 200) and 8 (adapted to the TI-Nspire CX II CAS) were released in 2023, accompanied by a comprehensive documentation rewrite whose worked examples—drawn from nine widely used textbooks—now number in the hundreds. Version 7 kept Frederiksen’s *DiffEq*; version 8 uses his Laplace functions as adapted for the TI-Nspire by Philippe Fortin.

## 1.7 Version 9: the computer-algebra port (2026)

Version 9, released in 2026, is a port of the classic TI-Basic code to Python, with the SymPy computer algebra library supplying the symbolic mathematics that the calculators’ built-in CAS supplied before. It runs anywhere Python runs—phones, tablets, laptops, notebooks, and, via WebAssembly, entirely inside a web browser—and, no longer bound by calculator memory, solves circuits faster than its predecessors. The port was carried out by the author working with Claude, Anthropic’s AI assistant, completing in weeks a task that had waited two decades. Symulator has always been free of charge; since 2026 it is open source under the MIT licence, published as the `symulator` package on PyPI.

This monograph documents the solver’s internal logic as a single conceptual algorithm. Where the calculator versions and the computer-algebra port genuinely differ in *mechanism* (though never in results), the difference is noted in place, and Chapter 3 collects the calculator-era mechanisms worth remembering.

## Chapter 2

# The logic of Symbulator

This chapter explains how Symbulator solves a circuit, using direct current as the template; Chapter 3 then presents every other analysis as a variation on this template. We proceed in words first, and in pseudo-code second.

### 2.1 The idea in words

Symbulator’s whole pipeline can be said in one sentence, and was, in the 1999 paper [1]: *the program reads the circuit description, writes down the equations that describe its behavior, solves them, and stores all the answers in variables with their respective names.* Everything else is the careful working-out of that sentence.

The thesis states the governing idea in terms this monograph will honor throughout [2, §1.4]: “To analyze an electric circuit, Symbulator uses a method that *emulates the manual circuit-analysis techniques that students learn in the classroom.* In fact, the simulator behaves like an excellent student who paid attention and learned the lesson. Since the program follows a path reproducible by the student, it is also a very powerful learning tool.” The same passage fixes the method’s austerity: Symbulator knows the Ohm–Cavendish law and Kirchhoff’s current law, and with those two laws alone it analyzes circuits using nodal analysis exclusively. It never uses mesh analysis; and “although it does not know Kirchhoff’s voltage law, it always satisfies it, because that law is satisfied automatically when the other two are.” KVL, in other words, is *emergent* in this architecture: every voltage is expressed from the start as a node potential (or a difference of two), so no loop sum can ever fail to close. A second principle governs the input side [2, §1.8.1]: “the user must provide the simulator with only the *minimum information necessary* to completely describe each element of the circuit”—a philosophy visible in every line of the description language of §2.2.

*Reading* means parsing a plain-text description in which each line names one element, its terminals, and its value—the value being a number, a symbol, or an expression; a dependent source is simply a source whose value names another answer. Reading ends with validation: the description must name a reference node, no element may have both its terminals on the same node, and every node must have a conduction path to the reference.

*Writing down the equations* is called *stamping*: one pass over the element list in which each element contributes its own defining voltage–current relationship (Ohm’s law for a resistor, a fixed voltage for a source), and adds its current into a running per-node sum. When the pass ends, each node’s sum is set to zero: by Kirchhoff’s Current Law (KCL), the currents leaving a node must balance. One equation per element law, one per non-reference node.

*Solving* hands the system to the platform’s equation solver—the TI-89’s `solve()` then, SymPy’s now. But not every answer needs to be *in* the system, and this is the architectural idea Symbulator has carried since 1999: answers have *levels*. First-level unknowns are the ones the solver must find simultaneously: the node voltages, and the currents nothing local determines—a voltage source fixes a voltage and supplies whatever current the circuit demands, so its current can only come out of the joint solve. Second-level answers are expressions in the first-level variables, written down rather than solved for: a resistor’s or

a capacitor’s current follows from the solved voltages by the element’s own law, by direct substitution; a current source’s current is the plainest expression of all—known outright, since it *is* the source’s value, entering its two nodes’ KCL sums as a plain term rather than through any equation. Third-level quantities are derived after the solve, by plain algebra on everything before them: the voltage drop across each element, the power it absorbs, and, for each source, the resistance the source sees looking into the rest of the circuit.

*Storing the answers* gives every quantity a systematic name —  $v_n$  for node voltages,  $i_e$  for currents,  $v_e$ ,  $p_e$ ,  $r_e$  for the derived quantities of element  $e$  — and it is by these names that dependent sources, expert-mode equations, and the user’s own follow-up questions refer to them.

The vocabulary and the doctrine are the thesis’s own: its theory chapter is devoted to the levels [2, §4.2]. The *first-level equations* are “the indispensable equations Symbulator uses to solve a circuit,” and come in exactly two kinds: one *node equation* per node—the KCL sum, with resistive currents written out by Ohm’s law—and one *special equation* per *special element*. The thesis is charmingly precise about what “special” means: since the analysis is nodal, “the only elements that are *not* special... are the resistor and the current source”; a voltage source and even a short circuit are special, each contributing one equation. The count is therefore knowable in advance—“for a circuit with 5 nodes, 2 voltage sources and 1 short circuit, Symbulator will generate a total of 8 first-level equations. It does not matter if this circuit has a thousand resistors and seven hundred current sources: it will still have 8 first-level equations”—and the *first-level variables* match them one for one: the node voltages, plus the currents of the special elements. Everything else is expressed, not solved: for each remaining answer the same pass generates a *second-level expression* “given exclusively in terms of first-level variables,” of which some (resistor and capacitor currents) are *evaluated* into numbers after the solve while others (current-source currents, voltage drops) are kept as standing expressions; and finally the *third-level variables*—the powers, in DC and AC only—are built from everything before them, “defined at the end by the very nature of power: the product of a voltage drop and a current” [2, §4.2.6]. The thesis even carries the census forward as new elements appear [2, §9.4.3]: an inductor’s current is first level; a capacitor’s is a second-level expression that is evaluated; an ideal transformer’s primary current is first level, its secondary a kept expression; both currents of a two-port are first level; an op-amp’s output current is first level.

Keeping the simultaneous set small was the point: it is what made symbolic solving feasible on a late-1990s calculator. Version 9, with SymPy underneath, solves the joined system in one call and keeps the staged view as what it always really was—the right way to *understand* the solver. Where this monograph classifies an answer, it classifies by what the version 9 code does, and twice that departs from the census above. Resistor currents, second-level expressions on the calculator, are stamped straight into the simultaneous solve (§4.5). And the voltage drops, which the thesis keeps as second-level standing expressions [2, §4.2.4], are computed by the port after the solve, in the round the code itself labels third level—beside the powers, and beside the resistance each source sees, an answer the census never reached at all.

## 2.2 The input language

### 2.2.1 Circuit descriptions

A circuit is described as plain text, one element per line (or colon-separated, in the calculator style). Each element is a comma-separated list beginning with its name; the first letter of the name selects the element’s type. Table 2.1 gives the complete catalog.

Table 2.1: Element catalog. The first letter of the name selects the type.

| Letter      | Element               | Fields after the name                   |
|-------------|-----------------------|-----------------------------------------|
| r           | resistor              | $n_1, n_2, \text{value}$                |
| l           | inductor              | $n_1, n_2, \text{value } [, i(0)]$      |
| c           | capacitor             | $n_1, n_2, \text{value } [, v(0)]$      |
| e           | voltage source        | $n_1, n_2, \text{value}$                |
| j           | current source        | $n_1, n_2, \text{value}$                |
| o           | ideal op-amp (nullor) | $n_+, n_-, n_{\text{out}}$              |
| m           | mutual inductance     | $L_{\text{name1}}, L_{\text{name2}}, M$ |
| s           | short circuit         | $n_1, n_2$                              |
| t           | ideal transformer     | $n_1, n_2, N_1, N_2$                    |
| z y h g a b | grounded two-port     | $n_1, n_2$                              |

Sources may be *independent* (a numeric or symbolic value) or *dependent*: a value may reference any other answer by name, so `j2,3,0,0.5*i_r1` is a current source of half the current through `r1`, and `e2,2,0,3*v_rx` a voltage source controlled by the drop across `rx`. This uniform treatment—a dependent source is just a source whose value happens to mention an answer—is one of Symulator’s simplest and most distinctive design decisions, and was among the reasons the equation-generation method was chosen in the first place: no special element types, no auxiliary defining branches, merely an expression with a name in it.

The optional trailing field on `l` and `c` is an initial condition (initial inductor current; initial capacitor voltage), read only by the FD and TR analyses and treated as zero when absent. `m` is unusual in that it couples two *elements* (named inductors, or, in AC, elements holding impedances) rather than two nodes. The six two-port letters denote a black box characterized only by its four `z/y/h/g/a/b` parameters, each port’s second terminal being implicitly ground.

### 2.2.2 Sign conventions

The description language carries one piece of information that is easy to miss and impossible to reimplement the solver without: the *order of the nodes is data*. The thesis states the convention as a single uniform rule [2, §2.11.1]: for every element, the current reported in the answers is defined as flowing *through the element from the first listed node to the second*, and the voltage drop as the potential of *the first listed node with respect to the second*. For a current source, the first-to-second direction must therefore match the arrow in the schematic; for a voltage source, the first node is the  $+$  terminal; for a resistor the order is free, chosen for convenience (typically to make a requested current come out with the expected sign); for coupled inductors the order encodes the dot convention (§3.1); for an initial condition it fixes the polarity of  $v(0)$  or the direction of  $i(0)$ .

Two corollaries the thesis spells out at length, because they are where beginners stumble. First, defining a polarity is not asserting a sign: “the voltage of node A is  $-5$  volts with respect to node B” defines the drop from A to B and simultaneously tells us B is the more positive—a negative answer is the convention doing its job, not an error. Second, *power is reported as consumed*: a source that delivers 80 W reports  $p = -80$  W. Both conventions are load-bearing in what follows—the KCL bookkeeping of Algorithm 2 adds each current at its first node and subtracts it at its second, and the third-level round of Algorithm 3 computes  $p = (v_{n_1} - v_{n_2}) i$  with no sign adjustment, precisely because the two conventions agree.

### 2.2.3 Answer names, and the spelling equivalence

Every quantity the solver produces is stored under a systematic name, and these names are part of the input language too: a dependent source’s value, an expert equation, and a user’s follow-up question all refer to answers by name. Table 2.2 gives the complete inventory. The scheme is the calculator’s, unchanged since 1999 except for one typographic reform: version 9 writes an underscore between the prefix and the element name (`i_r1` where the calculator wrote `ir1`), so that machine-generated output is unambiguous to read back.

Table 2.2: The complete answer inventory for a circuit element or node. Names on the left are version 9’s canonical spellings; the calculator’s flat spellings (right) remain valid everywhere.

| Answer              | Meaning                                     | v9 name                              | classic name     |
|---------------------|---------------------------------------------|--------------------------------------|------------------|
| node voltage        | potential of node $n$ vs. ground            | <code>v_n</code>                     | <code>vn</code>  |
| current             | through element $e$ , $n_1 \rightarrow n_2$ | <code>i_e</code>                     | <code>ie</code>  |
| voltage drop        | across $e$ , $n_1$ vs. $n_2$                | <code>v_e</code>                     | <code>ve</code>  |
| power (DC)          | absorbed by $e$                             | <code>p_e</code>                     | <code>pe</code>  |
| complex power (AC)  | absorbed by $e$                             | <code>s_e</code> / <code>ap_e</code> | <code>se</code>  |
| seen resistance     | looking out from source $e$                 | <code>r_e</code>                     | <code>re</code>  |
| seen impedance (AC) | looking out from source $e$                 | <code>z_e</code>                     | <code>ze</code>  |
| transformer current | into winding at node $n$                    | <code>i_tn</code>                    | <code>itn</code> |
| two-port current    | into port at node $n$                       | <code>i_pn</code>                    | <code>ipn</code> |

Since solver 0.5.19, the two spelling traditions are formally *one namespace*. Each circuit builds an alias map from its own inventory—its nodes and elements—and any name that *normalizes* to a known answer name (comparison is case-insensitive and ignores underscores) is canonicalized to it wherever expressions are read: element values, expert equations, unknowns, and conditions alike. `ir1`, `i_r1`, `IR1`, and `I_R1` are the same current, exactly as they were one calculator variable in 1999; a netlist from the thesis runs verbatim today (§5.1). Names that match nothing in the circuit are left untouched and pass through as free symbols, so the design keeps its other virtue: a symbol the user invents is never captured by accident, only spellings of answers the circuit actually has.

### 2.2.4 The value language

Element values, source waveforms, expert-mode equations, and tool arguments all pass through one expression reader, which accepts:

- **Numbers and symbols**, and arithmetic over them (`+` `-` `*` `/`, exponentiation as `^` or `**`, and function calls such as `sqrt`, `exp`, `sin`). Implicit multiplication is inserted where the calculator allowed it (`2ir3`  $\rightarrow$  `2*ir3`), and `e^x` is read as  $e^x$ . Any name is an ordinary symbol—including, since version 0.5.19, names that happen to be reserved words of the host language, such as `is`: the reader’s language is circuit analysis, not Python, and its natural names all work. Names that spell an answer of the circuit, in either the underscored or the flat tradition, resolve to that answer (§2.2.3).
- **SI prefixes** in the calculator’s quoted notation: `4.7'u` is  $4.7 \times 10^{-6}$ , `1'k` is  $10^3$ . A *bare* suffix (`1k`) is inherently ambiguous—one thousand, or one times a variable named  $k$ ?—so a policy argument decides: refuse and ask (the default), read all as SI, or read all as products.

- **Phasors** in textbook angle notation:  $(110\angle-120^\circ)$ . These are converted to rectangular complex numbers on entry. The conversion is deliberate: carrying  $e^{j\theta}$  factors symbolically makes some mesh systems practically unsolvable, while the rectangular form solves in seconds; the cost is exactness of the angle, accepted on the view that a phasor angle is a measurement.
- **Waveform shorthands** for TR sources:  $\mathbf{u}(\mathbf{t})$  is the unit step (Heaviside),  $\mathbf{delta}(\mathbf{t})$  the unit impulse (Dirac).
- **The parallel shortcut**  $[\mathbf{a}, \mathbf{b}, \dots]$ , expanded to  $\text{pr}(a, b, \dots)$ , the parallel combination  $(\sum_k 1/z_k)^{-1}$  with the convention that a zero branch dominates ( $\text{pr} = 0$ ).
- **Braces**  $\{\text{expr}\}$  in FD source values, meaning “this value is written in the time domain; transform it,” i.e.  $\mathcal{L}\{\text{expr}\}$ .

Because descriptions may come from strangers over the web, the reader is also a security boundary: expressions are parsed into a syntax tree and refused unless they are plain arithmetic, and names resolve against a small fixed namespace (so a user’s  $\mathbf{Q}$  is a symbol, never an internal object of the algebra system).

## 2.2.5 Validation

Before any equation is written, the parsed element list is validated (Algorithm 1). Names and node labels are case-insensitive and fold to lower case; duplicate names are refused. Each element must have exactly the field count its type demands (one more is allowed on  $1/\mathbf{c}$ , for the initial condition). The whole-circuit checks then enforce three properties that the solver depends on: a reference node exists; no element has both its terminals on the same node—a rule that binds even the short circuit, whose job is to join two *distinct* nodes: a true self-loop would leave its own current indeterminate, entering and leaving the same KCL sum; and every node has a conduction path to the reference. The last check matters more than it may appear: a floating sub-circuit has no defined voltages, and without the check the solver would quietly return its voltages parametrized in one another ( $v_2 = v_3$ ) rather than flag the mistake.

## 2.3 Notation for the pseudo-code

Throughout,  $n_1, n_2$  denote an element’s two terminals as written in the description, node 0 is the reference (ground),  $v_n$  is the voltage of node  $n$  with  $v_0 \equiv 0$ , and  $i_e$  is the current through element  $e$ , defined positive when flowing from  $n_1$  to  $n_2$  *through the element*. In pseudo-code,  $\mathcal{E}$  is the parsed element list,  $\Sigma$  the equation system under construction,  $\mathcal{U}$  the list of unknowns, and  $\mathcal{K}$  the map of *known* quantities (named answers that never become unknowns). Assignment is  $\leftarrow$ ; EMIT adds an equation to  $\Sigma$ ; and “adds  $i$  at  $n_1$ ,  $-i$  at  $n_2$ ” abbreviates the KCL bookkeeping  $\text{sum}[n_1] += i$ ;  $\text{sum}[n_2] -= i$ .

## 2.4 DC analysis, annotated

We now walk the full DC pipeline in pseudo-code. Everything in this section is the template; the other analyses (Chapter 3) change only the per-element stamping rules and the transforms around the solve.

A word on ambition before we begin. This section, together with the element catalog (Table 2.1), the sign conventions (§2.2.2), the answer inventory (Table 2.2), and the per-domain stamping rules of Chapter 3, is intended as a *complete* specification: a reader who

---

**Algorithm 1** PARSE-AND-VALIDATE: description text to element list

---

**Input:** description text  $D$ **Output:** element list  $\mathcal{E}$ , or an error naming the problem

```
1: split  $D$  into element strings (on newlines or colons)
2: for all element string  $x$  do
3:   expand notation shorthands in  $x$  ▷  $\angle$ ,  $[\ ]$ , quoted SI prefixes...
4:   split  $x$  on top-level commas into name and fields
5:   fold name and node fields to lower case
6:   check first letter of name is a known type
7:   check name not already used; field count fits type
8:   append Element(name, kind, fields) to  $\mathcal{E}$ 
9: end for
10: check some element touches node 0 ▷ a reference exists
11: for all  $e \in \mathcal{E}$  with two terminals do
12:   check  $n_1 \neq n_2$  ▷ no self-loop: both terminals on one node
13: end for
14: union-find over terminals: each element joins its nodes ▷ op-amps join all three;
   grounded two-ports join theirs to 0
15: check every node is connected to 0 ▷ no floating sub-circuit
16: return  $\mathcal{E}$ 
```

---

works through it with care should be able to implement a symbolic circuit simulator of their own, needing from their platform only what Symbolator has always needed from its own—an equation solver (`solve()` on the TI-89; `sympy.solve` today), and, for the transient analysis, a Laplace transform pair. That was the test the thesis set itself in 2001, teaching the internals well enough that an expert user could take over the machinery mid-flight (§3.5); it is the test this monograph sets itself now.

### 2.4.1 Stamping

Algorithm 2 is the heart of the solver. Two bookkeeping conventions do quiet but important work. First, asking for a node’s voltage ( $v(n)$  in the pseudo-code) is what *creates* the node: it registers  $v_n$  as a first-level unknown and opens the node’s KCL sum, so even a node touched only through a voltage relation (an op-amp input, say) still ends the pass with a KCL row. Second, ground never gets a symbol:  $v(0)$  is the constant 0, so the reference is built into every equation rather than solved for.

A word on each rule, in the order stamped. The *resistor* is the paradigm: one law, one new second-level current, two KCL contributions. The *inductor* at DC steady state sustains no voltage—it is a short—but its current is a real answer and stays in the system. The *capacitor* at DC steady state carries no current—it is an open—so its current is recorded as the constant 0 in  $\mathcal{K}$  and nothing enters the system at all. The *voltage source* fixes a voltage and leaves its current free; the *current source* fixes a current and leaves its voltage free, and because the fixed thing *is* the answer, nothing about it is ever solved: the current source is, with the resistor, one of the thesis’s two non-special elements, contributing no equation and no unknown. The *op-amp* is modeled as a nullor: the two inputs are forced equal and draw no current—their absence from the KCL sums *is* the infinite input impedance—while the output contributes a free current with no return path in the model, standing for the supply rails. The *ideal transformer* relates its winding voltages by the turns ratio and its currents by the inverse ratio, so that it neither creates nor consumes power; only one winding current is a free unknown. A *two-port block* is a black box of four parameters (given numerically, or

---

**Algorithm 2** STAMP-DC: element list to equation system (DC rules)

---

**Input:** validated element list  $\mathcal{E}$ **Output:** system  $\Sigma$ , unknowns  $\mathcal{U}$ , knowns  $\mathcal{K}$ 

```
1:  $\Sigma \leftarrow \emptyset$ ;  $\mathcal{U} \leftarrow \emptyset$ ;  $\mathcal{K} \leftarrow \emptyset$ ;  $\text{sum}[n] \leftarrow 0$  for all  $n$ 
2: for all  $e \in \mathcal{E}$  do
3:   switch  $\text{kind}(e)$  do
4:     case r (resistor, value  $R$ ):
5:       if  $R = 0$  then treat as s below ▷ a  $0\Omega$  resistor is a wire
6:       else
7:         EMIT  $v_{n_1} - v_{n_2} = R i_e$ ; add  $i_e$  to  $\mathcal{U}$  ▷ Ohm's law;  $i_e$  is second level
8:         adds  $i_e$  at  $n_1$ ,  $-i_e$  at  $n_2$ 
9:       end if
10:    case l (inductor):
11:      EMIT  $v_{n_1} - v_{n_2} = 0$ ; add  $i_e$  to  $\mathcal{U}$  ▷ a short at DC steady state; its current still
12:      adds  $i_e$  at  $n_1$ ,  $-i_e$  at  $n_2$  matters
13:    case c (capacitor):
14:       $\mathcal{K}[i_e] \leftarrow 0$  ▷ open at DC steady state: no equation, no unknown
15:    case e (voltage source, value  $E$ ):
16:      if  $E = 0$  then treat as s below
17:      else
18:        EMIT  $v_{n_1} - v_{n_2} = E$ ; add  $i_e$  to  $\mathcal{U}$  ▷ + at  $n_1$ ; current left free—the source
19:        adds  $i_e$  at  $n_1$ ,  $-i_e$  at  $n_2$  supplies whatever is demanded
20:      end if
21:    case j (current source, value  $J$ ):
22:       $\mathcal{K}[i_e] \leftarrow J$  ▷ second level: the current is the value
23:      adds  $J$  at  $n_1$ ,  $-J$  at  $n_2$  ▷ no equation, no unknown; the voltage across is free
24:    case o (op-amp, terminals  $n_+, n_-, n_{\text{out}}$ ):
25:      EMIT  $v_{n_+} = v_{n_-}$ ; add  $i_e$  to  $\mathcal{U}$  ▷ the nullor: inputs equal, drawing no current
26:      adds  $-i_e$  at  $n_{\text{out}}$  only ▷ output supplies what balances its node; the missing
27:      return path is the supply rails
28:    case s (short circuit):
29:      EMIT  $v_{n_1} - v_{n_2} = 0$ ; add  $i_e$  to  $\mathcal{U}$ ; adds  $i_e$  at  $n_1$ ,  $-i_e$  at  $n_2$ 
30:    case t (ideal transformer, turns  $N_1, N_2$ ):
31:      EMIT  $v_{n_1}/N_1 = v_{n_2}/N_2$ ; add  $i_1$  to  $\mathcal{U}$  ▷ voltages by the turns ratio
32:      adds  $i_1$  at  $n_1$  and  $-i_1 N_1/N_2$  at  $n_2$  ▷ currents by the inverse ratio: power in =
33:      power out
34:    case z y h g a b (two-port, parameters  $p_{11..22}$ ):
35:      solve the family's two defining equations for  $(i_1, i_2)$  in terms of  $(v_{n_1}, v_{n_2})$  ▷ e.g.
36:      for z: invert the  $2 \times 2$  matrix
37:      add  $i_1, i_2$  to  $\mathcal{U}$ ; EMIT their defining equations; adds  $i_1$  at  $n_1$ ,  $i_2$  at  $n_2$ 
38:    end switch
39:  end for
40: for all registered node  $n \neq 0$  do
41:   EMIT  $\text{sum}[n] = 0$  ▷ KCL: one equation per non-reference node
42: end for
43: CLOSE-KNOWN-REFERENCES( $\Sigma, \mathcal{K}$ ) ▷ §2.4.2
```

---

left symbolic to be referenced or solved for); whatever the family, its two defining equations are rearranged at stamp time into the form KCL wants—port currents in terms of port voltages. Finally, every *zero-valued* voltage-defining element ( $0\ \Omega$ ,  $0\ \text{H}$ ,  $0\ \text{V}$ ) collapses to the short-circuit rule; a  $0\ \text{F}$  capacitor, by contrast, is already an open circuit under the capacitor rule, with no special case needed.

At DC, a mutual inductance contributes nothing: its induced-voltage term rides on  $j\omega$  (or  $s$ ), which is zero here, and the classic versions simply ignored the element in this analysis [2, §6.11], as version 9 effectively does through the inductor’s short-circuit stamp. The classic versions ignored the ideal transformer at DC as well [2, §6.12]; version 9 stamps its ratio equations in every domain, since they contain no frequency term to vanish.

### 2.4.2 Closing references to known answers

A dependent source may name a quantity that is *known* rather than solved for: another source’s current, a capacitor’s current, or an element’s voltage drop (which is never an unknown—it is derived after the solve, as  $v_{n_1} - v_{n_2}$ ). Left alone, such a name would enter the system as a fresh free variable that no equation constrains, and the solver would happily answer every quantity *in terms of it*—a failure disguised as a solution. The final stamping step therefore substitutes, everywhere in  $\Sigma$ , each known name by its stamped expression, iterating to a fixed point since one known may name another (a current source controlled by a capacitor’s current is exactly that). Node-voltage names win where a node and an element share a name. After this closure, a reference like `0.5*i_cx` has become a real constraint on node voltages.

### 2.4.3 Solving by levels

Algorithm 3 completes the analysis. On the calculators, the simultaneous system of line 2 contained only the unknowns that truly required it (the first *level*), everything locally computable being deferred to the substitution loop; in version 9 the split survives only for the two always-local cases (current-source and capacitor currents), and line 2 is a single call to `sympy.solve`. Either way the third-level round is the same plain algebra: drops, powers, and—for sources only, where the question “what does this source see?” is meaningful—the seen resistance. The answers are stored under the systematic names of Table 2.2, mirroring the calculator variables (`v2`, `ir1`, `pe`, `re`) under which the original stored them—and this store of named, reusable answers was the program’s declared output philosophy from the start: “give the user, as output, all the answers they could desire,” shown only on request so as not to overwhelm [2, §2.5.1].

### 2.4.4 Diagnosing the unsolvable

When the solve comes back empty, the solver does not merely apologize: it looks for the two classic contradictions of linear circuit theory and names the culprit elements. A cycle in the graph of voltage-defining edges—voltage sources, shorts, zero-ohm resistors, and, in DC, inductors—pins the same voltage more than once, and the offending loop is found by union-find and reported by name (“two sources in parallel,” in the textbook phrase). Dually, a node touched only by current-defining elements—current sources and, in DC, capacitors—has its KCL fixed by constants that cannot balance, and is reported likewise. Only if neither pattern is present does a generic message stand.

---

**Algorithm 3** SOLVE-DC: the staged solution

---

**Input:**  $\Sigma, \mathcal{U}, \mathcal{K}$  from STAMP-DC; expert extras if any (§3.5)

**Output:** ranked list of answer maps  $\{\text{name} \mapsto \text{expression}\}$

```
1: EXPERT-AUGMENT( $\Sigma, \mathcal{U}, \mathcal{K}$ ) ▷ Alg. 5, if extras given
   — the simultaneous system: first level (and, in version 9, most of the second) —
2:  $\mathcal{S} \leftarrow \text{solve}(\Sigma, \mathcal{U})$  ▷ all solutions; a plain circuit has exactly one
3: if  $\mathcal{S} = \emptyset$  then
4:   DIAGNOSE-UNSOLVABLE( $\mathcal{E}$ ) ▷ name the contradiction; §2.4.4
5: end if
6:  $\mathcal{S} \leftarrow \text{RANK}(\mathcal{S})$  ▷ §3.6; a single solution is untouched
7: for all solution  $\sigma \in \mathcal{S}$  do
   — the rest of the second level: direct substitution —
8:   for all  $(\text{name} \mapsto \text{expr}) \in \mathcal{K}$  do
9:      $\sigma[\text{name}] \leftarrow \text{expr}$  with  $\sigma$  substituted in ▷ knowns were stamped in terms of  $v_n$ 
     symbols
10:   end for
   — third level: drops, powers, seen resistances —
11:   for all  $e \in \mathcal{E}$  of kind e j r c l do
12:      $\Delta v \leftarrow \sigma[v_{n_1}] - \sigma[v_{n_2}]; \quad i \leftarrow \sigma[i_e]$ 
13:      $\sigma[v\_e] \leftarrow \Delta v$  ▷ voltage drop across  $e$ 
14:      $\sigma[p\_e] \leftarrow \Delta v \cdot i$  ▷ power absorbed by  $e$ 
15:     if  $e$  is a source (e or j) then
16:        $\sigma[r\_e] \leftarrow \Delta v / (-i)$  ▷ resistance the source sees looking out
17:     end if
18:   end for
19:   for all op-amp  $e$  do
20:      $\sigma[p\_e] \leftarrow \sigma[v_{n_{\text{out}}}] \cdot (-\sigma[i_e])$  ▷ power at the output node
21:   end for
22: end for
23: return  $\mathcal{S}$ 
```

---

# Chapter 3

## Variations on the template

Every other analysis, and every mode, is the DC template of Chapter 2 with a controlled variation: different stamping rules (AC, FD), transforms wrapped around the solve (TR), extra equations folded into the system (expert mode), or—in the calculator era—optimizations that kept the symbolic load within a handheld’s reach.

### 3.1 Alternating current

AC analysis works in phasors at a single angular frequency  $\omega$  (a number or a symbol). Three stamping rules change, and the third level grows richer:

- The *inductor* becomes an impedance:  $v_{n_1} - v_{n_2} = j\omega(L i_e + \sum_{e'} M_{ee'} i_{e'})$ , the sum ranging over mutually-coupled partners—one coil’s current induces voltage in the other, which is what magnetic coupling means. A reversed winding is a negative  $M$ .
- The *capacitor*’s current becomes locally computable again, now nonzero:  $i_e = j\omega C (v_{n_1} - v_{n_2})$ , recorded in  $\mathcal{K}$  exactly as in DC. When  $C = 0$  this evaluates to an open circuit on its own, with no special case.
- A *resistor* carrying a complex value is an impedance in ohms; when such elements are coupled by an  $\mathbf{m}$  whose value is also an impedance, the mutual term  $M i_{e'}$  enters with no  $j\omega$  factor—the value already *is* an impedance. This matches how textbooks give coupled pairs “in  $j\Omega$ .”

At the third level, complex power replaces DC power:  $S_e = V_e \overline{I_e}$  under RMS phasors, or  $V_e \overline{I_e}/2$  under amplitude phasors, with the real part reported as the average power (named  $\mathbf{p}$  in the first convention and  $\mathbf{ap}$  in the second, to keep the distinction audible), and sources also receive the impedance  $z_e$  they see. The names  $i, j$  are reserved as the imaginary unit only here—in every other domain they remain free for the user.

The two conventions are themselves an inheritance. The thesis teaches both as the *rigorous* and the *informal* treatments of RMS problems [2, §6.13]: rigorously, sources are entered at amplitude and every answer divided down ( $\sqrt{2}$  for RMS voltages and currents, 2 for average powers); informally, one enters every independent source already in RMS, whereupon—by linearity—“all the voltages and currents of the answers will be in RMS, and all the powers in average,” with no division at all. Version 9 keeps both roads open by naming the two power conventions differently, so a description whose sources are RMS phasors reads its  $\mathbf{p}$  answers as averages directly, exactly as the informal treatment promises.

### 3.2 Frequency domain

FD analysis stamps in the Laplace variable  $s$ , and the energy-storage elements carry their initial conditions into the equations:

$$\text{inductor: } \frac{v_{n_1} - v_{n_2}}{s} = L \left( i_e - \frac{i_e(0)}{s} \right) + \sum_{e'} M_{ee'} \left( i_{e'} - \frac{i_{e'}(0)}{s} \right), \quad (3.1)$$

$$\text{capacitor: } i_e = sC (v_{n_1} - v_{n_2}) - C v_e(0), \quad (3.2)$$

the transforms of  $v = L di/dt$  and  $i = C dv/dt$  with the initial state folded in. The initial condition rides in the element’s fifth field, its polarity fixed by the node order (§2.2.2), and is taken as zero when absent—all three conventions stated in the thesis exactly as they stand today [2, §10.3]. Source values are expected already written in  $s$  (a step of amplitude  $k$  is  $k/s$ ), with the brace shorthand `{expr}` transforming a time-domain value on entry. No third level is computed in FD: an  $s$ -domain power has no established meaning in the curriculum Symulator serves, and the original made the same choice [2, §10.4].

### 3.3 Transient analysis: a Laplace round trip

TR is FD wrapped in two transforms (Algorithm 4): sources are carried *into* the  $s$ -domain, the FD engine solves, and every answer is carried *back* to time. The thesis presents the relationship candidly as the architecture it is [2, §11.1–11.2]: TR “solves the circuit in the frequency domain... and finally converts all the answers back to the time domain,” and the user is offered the two halves separately when that is cheaper—run `fd` and convert only the answers of interest (through the `fd_to_tr` helper, a thin wrapper on the inverse transform), or run `tr` and receive everything in  $t$  at the cost of inverting every answer. On the calculator the transform pair came from Lars Frederiksen’s *DiffEq*—in the thesis’s estimation “the fastest Laplace transform program ever written for a calculator”—invoked at the start of every FD or TR run and at the end of every TR run [2, §11.1.1]; version 9 calls SymPy’s pair at the same two seams. Both crossings have case analyses that are easy to get wrong and are worth stating exactly.

---

**Algorithm 4** TR: transient analysis via the  $s$ -domain

---

**Input:** description  $D$  with time-domain sources; expert extras

**Output:** time-domain answer map

---

— *sources in: for each independent source value  $x$*  —

- 1: **if**  $x$  mentions  $t$  **then**  $x \leftarrow \mathcal{L}\{x\}$  ▷ a waveform
- 2: **else if**  $x$  mentions  $s$  **then** leave  $x$  ▷ already transformed by the user
- 3: **else if**  $x$  references another answer **then** leave  $x$  ▷ a controlled source: a relation, not a waveform
- 4: **else**  $x \leftarrow x/s$  ▷ a constant: a step of that amplitude
- 5: **end if**
- 6: expert equations and conditions that reference answers cross the same way, side by side
- 7:  $\sigma \leftarrow$  FD solve of the converted description ▷ initial conditions live in the  $1/c$  stamps  
— *answers out: for each solved value  $y$*  —
- 8: **if**  $y$  mentions  $s$  **then**  $\sigma[y] \leftarrow \mathcal{L}^{-1}\{y\}$ , folding away a bare  $u(t)$  factor
- 9: **else if**  $y$  is a solved expert unknown **then** leave  $y$  ▷ a scalar, e.g. an amplitude
- 10: **else if**  $y$  references another answer **then** leave  $y$  ▷ reads identically in  $s$  and  $t$
- 11: **else**  $\sigma[y] \leftarrow y \cdot \delta(t)$  ▷ constant in  $s$  is an impulse
- 12: **end if**

---

On the way in, the four-way test distinguishes a waveform (transform it), a value the user already wrote in  $s$  (leave it), a controlled source (its value is a *relation* between answers—“twice whatever that current is”—which reads identically in either domain and must not be transformed), and a bare constant (a step). On the way out, an answer still carrying  $s$  is inverse-transformed, with one cosmetic rule: a factor of  $u(t)$  multiplying the whole answer is folded away, since every TR answer exists only for  $t \geq 0$  anyway—but only the exact factor  $u(t)$ ; a delayed step  $u(t - t_0)$  is genuinely informative and stays. An answer *constant* in  $s$  requires the most care: a step arrives as  $k/s$  and any waveform brings its own  $s$ , so a bare constant has nowhere else to come from than an impulse, and is reported as  $k \delta(t)$ —unless it is a solved expert scalar (an amplitude is a number, not a waveform) or a controlled source’s echo of its controlling answer. Answers whose inverse transform has no closed form are omitted individually rather than failing the analysis, since the rest of the circuit’s answers remain valid and useful.

One doctrine completes the transient picture, and the thesis states it as a rule of method rather than of code [2, §11.10]: *each interval of time requires its own simulation*. A circuit with a switch that acts at  $t = 0$  is two circuits—one before, one after—and the answers of the first supply the initial conditions of the second: typically a DC run for  $t < 0$  (capacitors open, inductors shorted, the steady state emerging automatically) whose node voltages and inductor currents are typed into the fifth fields of the  $t > 0$  description. Even  $u(-t)$  in a source value is to be read this way—“for the purposes of simulation,  $u(t)$  does not have that [mathematical] meaning; one must perform one simulation per interval”—the step function marking where one circuit ends and the next begins, not a value the solver should juggle across the boundary. The Boulet exemplar of §5.3 is this doctrine practiced exactly as taught.

### 3.4 The Impala mode

Time-dependent sources posed a particular threat on the calculators: a value like  $10 e^{-t} \sin(2t + \pi/6)$ , carried symbolically through every equation of a transient solve, could stall the CAS. The remedy, affectionately documented as the (*Impala*) mode [2, §11.11], exploits linearity. On entry, each complicated source value is replaced by a fresh placeholder symbol—the thesis specifies the scheme: a variable  $\sigma_{name}$ , its Greek initial marking it as one of the simulator’s internal variables, which all begin with Greek letters precisely so they can never collide with a user’s own [2, §1.8.4]. The system, now small and clean, is solved; and before the answers are stored, the Laplace transform of the original value is substituted for the placeholder *in every answer*. Because the circuit is linear, every answer is linear in the source values, and the late substitution is exact. Unlike the (Expert) mode, which the user must invoke, the (Impala) wakes by itself whenever such a source is present; and the thesis records the origin of the name: the African antelope that, threatened, escapes by prodigious leaps—“so too Symulator, on meeting very complex sources, leaps: it avoids using the complicated value and solves the equations with a simple symbolic one; then it lands, replacing the source by its complicated value in the frequency domain” [2, §11.11].

Version 9 no longer needs the mechanism as an optimization: SymPy carries the expressions directly. But the idea—abstract the messy constants, solve, substitute back—remains a fine trick for any symbolic-computation setting, and its trace survives in how the modern TR pipeline still classifies each source value before deciding what, if anything, to transform (Algorithm 4).

### 3.5 Expert mode

Expert mode extends a circuit with user-supplied mathematics, and is where Symulator turns from a calculator of answers into a design tool: instead of asking “what are the voltages, given every value?” one may fix an answer and solve for a value. Three extras are accepted, mirroring the prompts the calculator versions used:

- **Equations:** strings like `v_2 = 6` or `p_r1 = 0.025`, joined into  $\Sigma$  before solving;
- **Unknowns:** names to solve for—necessary when the symbol appears only in the *circuit* (a resistor of value `rx`), since the solver has no other way to know the symbol is to be varied rather than carried;
- **Conditions:** the calculator’s “|” (with) operator, in both of its meanings. An *equality* (`r_a = 1000`) is a substitution applied to the whole system at solve time—useful for trying numbers in an otherwise symbolic circuit. An *inequality* (`vs > 0`) is a restriction on which solutions may be returned, applied as a filter after the solve: exactly what `solve(...)|vs>0` did on the TI, and the natural way to select among the sign-symmetric roots a quadratic power constraint produces (see §5.2 for it in action). A restriction that excludes *every* solution is reported as such—the system solves, but the mathematics and the restriction disagree, which is an answer rather than a failure.

The original (*Expert*) mode was explicitly framed as opening the black box: in a normal simulation “the user only feeds in the circuit, while the entire responsibility for the simulation falls on the simulator”; the expert mode “gives the user the power to play a more important role, enabling them to modify the equations the simulator has generated, before the calculator solves them—in exchange, the user assumes part of the responsibility for the simulation” [2, §4.1]. Operationally, after the first-level equations were generated the calculator paused at a screen of four fields—the first-level unknowns, the first-level equations, and a pair of fields for *conditional* variables and their known values—plus a three-way menu: *Symbulate* (solve now), *Symbulate+Keep* (solve, and also store the system in variables `Equation`, `Unknown`, `Wheneq`, `Whenvar` for later reuse), or *Just Keep* (store without solving, to be solved later by the companion `solves()` tool) [2, §4.2.8–4.2.11].

The thesis then teaches, with worked examples, the two symmetric techniques that expert work reduces to. When a circuit carries one symbolic value and one known measurement, the generated system has  $n$  equations and  $n+1$  unknowns, and the expert may either *add an equation* (append `and ir3=4` to the system, and the symbolic value to the unknowns) or *eliminate an unknown* (declare a first-level variable’s known value in the conditional fields and remove it from the unknowns list). The trade-off is stated with an engineer’s economy: adding an equation is the easier road, but eliminating an unknown “has the advantage that the system of equations shrinks and can be solved more easily by the calculator—which matters for circuits much larger than those in these pages” [2, §4.3]. Two rules of legality guard the box’s edges: a *second-level* variable may appear in an added equation (its defining expression exists by then, so the name resolves), but only *first-level* variables may be declared in the conditional fields [2, §4.3.2]; and circuits with op-amps, or transient runs, must choose *Symbulate*—the former because their equations carry the internal limit term (§4.5), the latter because equations kept and solved later would yield answers still in the frequency domain [2, §9.4.1–9.4.2].

Version 9’s `equations/unknowns/conditions` arguments are these four fields, minus the screen: an expert equation is the “add an equation” technique, an equality condition is the “eliminate an unknown” technique (a substitution applied before the solve), and both may

name any answer of the circuit, first level or not, because the closure of §2.4.2 and the on-demand stamping below resolve whatever the name refers to. The restrictions fell away with the hardware: op-amps are stamped ideally rather than limited, and TR conversion happens inside the call.

---

**Algorithm 5** EXPERT-AUGMENT: fold user mathematics into the system

---

**Input:**  $\Sigma, \mathcal{U}, \mathcal{K}$ ; user equations  $Q$ , unknowns  $X$ , conditions  $C$

```

1: for all name  $x \in X$  do
2:   add symbol  $x$  to  $\mathcal{U}$  if not present
3: end for
4: for all equation  $q \in Q$  (parsed; bare expressions read as  $q = 0$ ) do
5:   substitute  $\mathcal{K}$  into  $q$  ▷  $i_{j1} = 5$  must constrain, not alias
6:   if  $q$  has no symbols left then
7:     if  $q$  is false then error “contradicts the circuit” else drop  $q$  ▷ tautology
8:   else
9:     EMIT  $q$ 
10:  end if
11: end for
12: for all new symbol  $x$  appearing in  $Q$  but not in  $\mathcal{U}$  do
13:   if  $x$  names a third-level quantity of some element then
14:     EMIT its defining equation; add  $x$  to  $\mathcal{U}$  ▷ see below
15:   else
16:     add  $x$  to  $\mathcal{U}$  ▷ a fresh derived quantity, e.g.  $p_{out}$ 
17:   end if
18: end for
19: for all condition  $x = v$  in  $C$ , in order do
20:    $\Sigma \leftarrow \Sigma[x := v]$ ;  $\mathcal{K} \leftarrow \mathcal{K}[x := v]$ ; remove  $x$  from  $\mathcal{U}$ 
21: end for

```

---

Algorithm 5 shows the folding, whose subtleties all stem from one theme: *an expert equation must constrain the circuit, never a lookalike variable*. Three cases require care.

First, an equation naming a known ( $i_{j1} = 5$  on a 5 A source) would otherwise introduce a free variable that shares the name and satisfies itself; substituting  $\mathcal{K}$  first turns it into a real constraint—or into a symbol-free residue, which is either a harmless tautology (dropped) or a flat contradiction (reported plainly).

Second, an equation naming a *third-level* quantity ( $r_e = 12000$ : “the source must see 12 k $\Omega$ ”) is the mirror problem: those names are computed after the solve and are invisible to it. The remedy is to stamp the quantity’s defining equation into the system on demand—for the seen resistance, in the polynomial form  $r_e \cdot (-i_e) = v_{n_1} - v_{n_2}$  so that a zero current cannot inject a division by zero—after which the constraint does real work. The definitions used here and the third-level round of Algorithm 3 are the same formulas by construction, and the test suite holds them to that.

Third, symbols appearing only in expert equations are adopted as unknowns automatically (a convenience the calculator did not offer), so that defining a new quantity,  $p_{out} = v_2 * i_{r2}$ , needs no separate declaration; an explicit unknowns list always wins over this inference.

### 3.6 Multiple solutions and their ranking

A circuit description alone yields a linear system with exactly one solution. Expert mode can change that: an added equation on a *power* is quadratic in the underlying unknowns, so a constraint like  $p_{r_1} = 0.025$  on a circuit with a symbolic source may be met by  $e = 5$  and  $e = -5$  alike. Symbulator returns *all* solutions, ranked so the physically likely one leads. The ranking judges only the *design* unknowns—symbols that are not node voltages or branch currents, e.g. a component value being solved for—since voltages and currents are legitimately negative all the time. Solutions whose design unknowns are all real rank above any with complex values; among those, all-non-negative ranks above any negative; ties preserve the algebra system’s order. The first solution is presented as *the* answer, with the alternatives available to inspect.

# Chapter 4

## Tools on top of the template

The equivalence tools contain no new circuit physics. Each one *orchestrates* the solver of the previous chapters: it appends one or two *test* elements to a copy of the description, re-runs the ordinary analysis, and reads the desired figure off the answers. This design—tools as thin clients of one trusted engine—is worth imitating: every tool automatically works in every domain the engine supports, with symbolic values, dependent sources, op-amps, and expert extras, because it is the same code path throughout.

### 4.1 Thévenin and Norton equivalents

---

**Algorithm 6** THEVENIN( $D, n_1, n_2, d$ ): equivalent seen from a port

---

```
1:  $\sigma_o \leftarrow$  solve  $D$  in domain  $d$  ▷ round 1: circuit as given (open port)
2:  $v_{th} \leftarrow \sigma_o[v_{n_1}] - \sigma_o[v_{n_2}]$ 
3: if  $v_{th} = 0$  then error “not active; use ER”
4: end if
5: try  $\sigma_s \leftarrow$  solve  $D + \mathbf{stest}, n_1, n_2$  in domain  $d$  ▷ round 2: port shorted
6:    $i_{no} \leftarrow \sigma_s[i_{stest}]$ 
7: on failure ▷ fallback: the short as a limit
8:   solve  $D + \mathbf{rtest}, n_1, n_2, x$ ;  $i_{no} \leftarrow \lim_{x \rightarrow 0^+} i_{rtest}$ 
9:   if the limit is unbounded then  $z \leftarrow 0$ ;  $p_{max} \leftarrow \infty$ ; return
10:  $z \leftarrow v_{th}/i_{no}$ 
11:  $p_{max} \leftarrow v_{th} i_{no}/4$  (dc)   or    $\frac{|v_{th}|^2}{8 \Re z}$  (ac, amplitude convention; 4 under RMS)
12: return ( $v_{th}, i_{no}, z, p_{max}$ )
```

---

The classic tool asked its user one question version 9’s does not: whether the network is *active* or *passive*. On Auto-Detect it ran two simulations regardless and reported  $v_{th}$ ,  $i_{no}$ ,  $z_{th}$ , and the maximum deliverable power ( $p_{max}$  in DC, complex  $s_{max}$  in AC); on Passive it saved a whole simulation—one round sufficed—at a price the thesis flags in italics: choose Passive on a network that is not passive and “the answers will be wrong. Never choose Passive if the network is not passive, or if in doubt” [2, §5.2]. The passive case survives in version 9 as the separate ER tool (§4.2); the active case is Algorithm 6. A definitional point the thesis makes twice, because students trip on it: a network containing only *dependent* sources is passive—such a network has a perfectly good equivalent resistance (possibly negative, as one thesis problem delights in showing with  $z_{th} = -r_x$ ) and no Thévenin voltage at all [2, §5.3, §9.3].

Algorithm 6 runs two rounds: the circuit as given yields the open-circuit voltage  $v_{th}$  between the chosen terminals, and a copy with a test short across them yields the short-circuit (Norton) current  $i_{no}$ ; then  $z = v_{th}/i_{no}$  and the matched-load maximum power follow. Two refinements matter in practice. If the shorted copy cannot be solved directly—which can happen when the short collapses part of the circuit degenerately—the tool falls back to

physics: a short is the limit of a vanishing resistance, so it plants a resistor of symbolic value  $x$  and takes  $x \rightarrow 0^+$ . An unbounded limit is then a real answer, not a failure: the port holds its voltage at any current, the equivalent impedance is zero, and  $p_{\max}$  is unbounded. Second, expert extras are passed to *both* rounds—correct for conditions and definitions, which mean the same thing in either round, though not for a measurement-style equation that holds only in the unshorted circuit, a caveat the tool documents.

## 4.2 Equivalent resistance of a passive network

For a source-free circuit, ER injects a 1 A test current source across the port and reports the resulting voltage—numerically equal to the equivalent resistance (DC) or impedance (AC):  $z_{\text{eq}} = (v_{n_1} - v_{n_2})|_{I_{\text{test}}=1}$ . One round, no ratios.

## 4.3 Two-port parameter extraction

---

**Algorithm 7** PORT( $D, n_1, n_2, \text{family}$ ): two-port parameters in one solve

---

- 1: pick test sources by family, with *symbolic* values  $x_1, x_2$ :
    - $z, a, b$ : current sources at both ports ▷ open-circuit family
    - $y$ : voltage sources at both ports ▷ short-circuit family
    - $h/g$ : current at one port, voltage at the other
  - 2:  $\sigma \leftarrow$  solve  $D +$  tests in domain  $d$  ▷ one round
  - 3: read  $V_1, V_2, I_1, I_2$  at the ports from  $\sigma$
  - 4: **for all** parameter  $p_{k\ell}$  of the family **do**
    - response at port  $k|_{x_{\bar{\ell}}=0}$
  - 5:  $p_{k\ell} \leftarrow \frac{\text{response at port } k|_{x_{\bar{\ell}}=0}}{x_{\ell}}$  ▷ zeroing  $x_{\bar{\ell}}$  opens/shorts that port
  - 6: **end for**
  - 7: for  $a/b$ : extract  $z$ -parameters as above, then convert by the closed-form  $z \rightarrow a$  ( $z \rightarrow b$ ) matrix identities
  - 8: **return**  $\{p_{11}, p_{12}, p_{21}, p_{22}\}$
- 

The classic PORT tool was the pride of the two-port chapter, and the thesis demonstrates it with a flourish worth retelling [2, §7.6–7.7]: given a bipuerto in  $z$  parameters, it extracts the  $y$  equivalent of *that two-port as a network*; then the  $h$  equivalent of the  $y$ ; then  $g$  of  $h$ ,  $a$  of  $g$ ,  $b$  of  $a$ —each extraction consuming the previous tool output as its input circuit—and finally, closing the loop, the  $z$  equivalent of the  $b$ , which lands exactly on the four numbers the problem began with. “Any error generated along the way would impact the final answer; obtaining the correct equivalent now means every intermediate equivalent was correct.” Six conversions through five parameter families, verified by round trip: the habit of *verification by re-derivation* that runs through the whole thesis, applied to the tool itself.

PORT (Algorithm 7) characterizes a circuit as a two-port of any of the six families. Its trick is to make the test sources’ values *symbolic*: with excitations  $x_1, x_2$  left as free symbols, a single solve captures the port responses as linear functions of both, and each parameter is read off by substituting the *other* port’s excitation to zero and dividing by this port’s own—algebraically identical to the textbook’s “open (or short) the other port and measure the ratio,” but with no second solve. The choice of source type per family mirrors the definitions: zeroing a test *current* opens a port, zeroing a test *voltage* shorts it, which is exactly the side condition each parameter family’s definition demands. The chain families  $a$  and  $b$  are obtained from an intermediate  $z$  extraction through their standard conversion identities rather than a bespoke excitation.

## 4.4 Small tools

Three helpers round out the toolkit, each with a history. `pr` is the parallel combination  $(\sum_k 1/z_k)^{-1}$  with a dominant-zero rule, available both as a function and as the `[a,b]` value shorthand; its ancestor `par` was written in first version by Erwin Baert of Belgium, one of the early users who became collaborators, and improved by the author [2, §5.4]—the thesis uses it to *derive* the parallel-resistor formula symbolically (`par(r1,r2) → r1r2/(r1 + r2)`) before ever using it numerically. `pf` reports the power factor  $|\cos(\angle V - \angle I)|$  of a voltage–current phasor pair, labelled *leading*, *lagging*, or *in phase* by the sign of the angle difference. `gain` computes, from input and output voltage–current pairs, the voltage gain  $A_v = v_2/v_1$ , current gain  $A_i = i_2/i_1$ , power gain  $A_p = \Re(-A_v \overline{A_i})$ , and input impedance  $Z_i = v_1/i_1$ ; the classic version was a four-field form whose answers (`av`, `ai`, `ap`, `zi`) were stored as variables like any others [2, §7.4].

Two display and plotting tools complete the classic inventory. `absang`—also Baert’s in origin, reworked by the author—printed any complex answer in polar form regardless of how the calculator’s display modes were set, the universal answer to “the book wants magnitude and angle” [2, §6.1.2]; version 9’s polar display option is its descendant. And the `bode` and `plot` tools turned an FD or TR answer into a graph on the calculator screen—the thesis demonstrates each on a problem from a professor at the author’s own university (§5.1, §5.3), and version 9’s plot menu covers the same ground with the host platform’s graphics.

## 4.5 What the port to version 9 changed

Version 9 is a faithful port: its stamping rules, naming conventions, level architecture, and tool behaviours were transcribed from the TI-Basic sources and validated against the printed answers of the version 7/8 documentation—322 worked examples, each run and compared. The genuine differences are few enough to enumerate, and each is instructive.

1. **Staging is now conceptual.** The level split was, on the calculator, an optimization that kept the simultaneous system within the hardware’s reach. SymPy removes the constraint, and the port solves the joined system at once, keeping direct substitution only where it is always locally computable (current-source and capacitor currents). Same physics, same answers, less machinery.
2. **Transforms come from the algebra system.** Versions 7/8 called Frederiksen’s Laplace libraries; version 9 uses SymPy’s `laplace_transform` and its inverse, with guards on both ends: a transform that comes back unevaluated is refused rather than propagated, and answers that fail to invert are omitted individually.
3. **One bug was fixed rather than preserved.** The original routed a 0 F capacitor through the zero-value short-circuit rule; a 0 F capacitor is an open circuit, and the port’s admittance-form stamping gets this right with no special case.
4. **Op-amps are stamped, not limited.** The earliest versions solved op-amp circuits with a finite model and applied the CAS `limit()` command to every stored answer [1]—idealization as a literal limiting process. The port stamps the ideal (nullor) equations directly.
5. **Phasors become rectangular numbers on entry.** The calculators’ CAS handled polar forms natively; SymPy carries  $e^{j\theta}$  unevaluated in ways that can stall a mesh system indefinitely, so angles are converted at the gate—trading a vanishing amount of exactness for solves that reliably terminate.

6. **Ambiguity is surfaced, not guessed.** A bare suffix (1k) meant SI on the calculator; in a library consumed by applications, it is refused by default with a structured error listing every ambiguous token, and the caller chooses a policy.
7. **Expert mode reaches further.** The original barred expert extras from the equivalence tools and required every new unknown to be declared; the port passes extras through the tools' rounds and adopts new symbols automatically, with the explicit list still winning.
8. **The two spelling traditions were reunified.** Version 9 introduced underscored answer names (`i_r1`) for machine-readability, which briefly split every answer's identity in two; solver 0.5.19 restored the calculator's reality by alias mapping (§2.2.3)—any spelling that normalizes to an answer of the circuit *is* that answer, in values, equations, unknowns, and conditions alike, so the thesis's own 2001 netlists run verbatim.

# Chapter 5

## Exemplars

Symbulator’s value shows best on circuits that are painful by hand and effortless once described. This chapter collects a few such exemplars, each solved by a single call—and two of them come with a rare kind of evidence: the thesis solved the same circuits on a TI-89 in 2001 and printed the answers [2], so the reader can watch the 2001 calculator and the 2026 port agree to the last digit. The schematic of every exemplar, drawn from its own description by version 9’s schematic engine, is collected in Appendix B.

### 5.1 The two-stage amplifier of 1999

The circuit that closed the original 1999 paper [1] returns in the thesis as Problem 87, the final course assignment set by Prof. Edilberto Yee of the Technological University of Panama [2, §12.2]: a two-stage small-signal amplifier model—source resistance, two RC loading networks, two voltage-controlled current sources ( $0.05 v_1$  and  $0.05 v_3$ ), coupling capacitors of picofarads against load resistors of kilohms—whose gain Bode diagram was requested over  $10^4$  to  $10^{13}$  rad/s. In Symbulator notation:

|                  |                                     |
|------------------|-------------------------------------|
| e1,5,0,vg        | <i>the source, behind...</i>        |
| r1,5,1,150       | <i>...its source resistance</i>     |
| r2,1,0,1'k       | <i>first stage: input loading,</i>  |
| cc1,1,0,100'p    | <i>...shunt capacitance,</i>        |
| cc2,1,2,3'p      | <i>...coupling capacitance</i>      |
| jd1,2,0,0.05*v_1 | <i>first stage's VCCS</i>           |
| r3,2,0,2'k       | <i>first stage's load</i>           |
| r4,2,3,100       | <i>interstage resistance</i>        |
| r5,3,0,1'k       | <i>second stage: input loading,</i> |
| cc3,3,0,100'p    | <i>...shunt capacitance,</i>        |
| cc4,3,4,3'p      | <i>...coupling capacitance</i>      |
| jd2,4,0,0.05*v_3 | <i>second stage's VCCS</i>          |
| r6,4,0,2'k       | <i>second stage's load</i>          |

(The circuit, as drawn by version 9’s own schematic engine, is Figure B.1 in Appendix B.)

An FD analysis returns every node voltage as a rational function of  $s$ ; the transfer function is one division,  $v_4/v_5$ , and the Bode plot follows. On the TI-89 this took about three minutes to solve and another minute to plot—against “around half an hour, or more” by hand, as any electrical engineering student of the era would attest [1]. On version 9 the same description solves in under a second. The thesis’s account of this problem ends on a note no benchmark can match: the plot “earned me full marks on the assigned homework—and, having obtained it with a simulator I had built myself, the respect of Professor Yee, whom I hold in the highest esteem” [2, §12.2].

The circuit makes a fine exemplar because it exercises so much at once: dependent sources written exactly as the textbook writes them, mixed magnitudes across nine orders, and a symbolic source riding through the whole analysis untouched. It also carries the spelling

equivalence of §2.2.3 as a live test: the thesis’s own description of this circuit is in version 9’s test suite element for element as printed in 2001—controls written  $5/100*v1$ , powers written  $10\sim 10$ , names run together, only the line separator modernised—and must keep producing the same transfer function for every release.

## 5.2 One of each: the 2013 showcase

The author composed this problem in 2013 with the express purpose of showcasing the powerful simplicity of Symbulator. The circuit contains one dependent source of each of the four species—a VCCS ( $0.2 v_{R7}$ ), a CCVS ( $0.1 i_{R5}$ ), a CCCS ( $2 i_{R1}$ ), and a VCVS ( $0.7 v_{R6}$ )—woven through seven resistors, with an independent voltage source  $V_s$  and an independent current source  $I_s$  left symbolic. The question inverts the usual direction of analysis:

*Find positive values of  $V_s$  and  $I_s$  such that the VCCS delivers 80 W and the CCVS dissipates 0 W.*

By hand this is a genuinely painful problem: two simultaneous *power* constraints—each quadratic in the underlying unknowns—on a network whose every mesh is entangled with a controlled source. In Symbulator it is a description and one expert call. Naming the nodes `a–n` and the elements after their labels:

|                               |                                        |
|-------------------------------|----------------------------------------|
| <code>es,e,0,vs</code>        | <i>the voltage source, symbolic</i>    |
| <code>js,0,d,is</code>        | <i>the current source, symbolic</i>    |
| <code>r1,e,m,10</code>        | <i>the seven resistors...</i>          |
| <code>r2,a,e,20</code>        |                                        |
| <code>r3,m,0,30</code>        |                                        |
| <code>r4,b,m,40</code>        |                                        |
| <code>r5,n,m,50</code>        | <i>...whose current the CCVS reads</i> |
| <code>r6,c,d,60</code>        | <i>...whose drop the VCVS reads</i>    |
| <code>r7,n,d,70</code>        | <i>...whose drop the VCCS reads</i>    |
| <code>jd1,a,b,0.2*v_r7</code> | <i>the VCCS (delivers the 80 W)</i>    |
| <code>ed2,c,b,0.1*i_r5</code> | <i>the CCVS (dissipates the 0 W)</i>   |
| <code>jd3,n,c,2*i_r1</code>   | <i>the CCCS</i>                        |
| <code>ed4,0,n,0.7*v_r6</code> | <i>the VCVS</i>                        |

with expert equations `p_jd1 = -80` (the solver’s *p* is power *consumed*, so delivering 80 W is consuming  $-80$ ) and `p_ed2 = 0`, unknowns `vs`, `is`,<sup>1</sup> and the positivity written directly as *conditions*: `is > 0`, `vs > 0`.

Every piece of Chapter 2 participates. The dependent values name answers on both sides of the solve:  $i_{R1}$  and  $i_{R5}$  are currents the system already carries as unknowns, while  $v_{R6}$  and  $v_{R7}$  are voltage drops—third-level names that the known-reference closure (§2.4.2) rewrites into node-voltage constraints. The expert equations name third-level powers, so their defining equations are stamped in on demand (§3.5)—and being powers, they are quadratic, so the system has *four* solutions: two sign-symmetric pairs. The inequality conditions act exactly

---

<sup>1</sup>The name `is` is worth a footnote of its own. When this chapter was first drafted, version 9 refused it: `is` is a reserved word of Python, and the port’s expression reader—which parses values through Python’s own grammar—let that refusal leak through to the circuits user, for whom `is` is the most natural name a source current could have. The drafting of this monograph exposed the wrinkle, and solver 0.5.19 removed it: Python keywords used as bare names are now shielded from the language’s parser and become ordinary symbols, so `is` works everywhere a name can appear.

as the calculator’s | operator did (§3.5): they filter the solve’s results, and the single call returns the one solution that survives:

$$V_s = \frac{3490}{\sqrt{39259}} \approx 17.614 \text{ V}, \quad I_s = \frac{551}{7\sqrt{39259}} \approx 0.39727 \text{ A}.$$

(Without the conditions, all four solutions come back, ranked with the all-positive pair first, per §3.6—either road reaches the answer; the conditions say what was meant.)

The solved circuit also explains *how* the zero-dissipation constraint is met, and the mechanism is elegant: in the winning solution  $i_{R5} = 0$  exactly. The C CVS’s voltage is  $0.1 i_{R5}$ , so the constraint drove its own controlling current to zero—the source dissipates nothing because the circuit has arranged for it to *be* nothing, a fact one reads directly off the answer sheet ( $i_{R5} = 0$ , hence  $v_{ed2} = 0$ , hence  $p_{ed2} = 0$ ) rather than divining it from the topology. This is the symbolic simulator’s characteristic gift: not just the numbers asked for, but the structure that produced them.

### 5.3 Prof. Eliane Boulet’s switching transient

This transient problem is a classic of the two-switch genre, set by Prof. Eliane Boulet de Cabrera in an examination at the Technological University of Panama—the same Prof. Boulet who would later sit on the thesis tribunal (§1.4)—and worked in the thesis as Problem 88, the closing problem of its closing chapter [2, §12.3]. For  $t < 0$ , an 18 V source behind  $8 \Omega$  feeds a resistive ladder ( $12 \Omega$ ,  $18 \Omega$ ,  $6 \Omega$ ) holding two capacitors ( $\frac{1}{6}$  F at node  $a$ , voltage  $v_1$ ;  $\frac{1}{3}$  F at node  $b$ , voltage  $v$ ); a waveform source  $10 e^{-t} \sin(2t + 30^\circ)$  A waits behind an open switch. At  $t = 0$  both switches act: the DC source disconnects, the waveform source connects. *Find  $v_1$  for  $t > 0$ .*

The problem is an exemplar because Symbulator solves *both* halves of it, and the halves chain. The  $t < 0$  circuit is a DC analysis—capacitors open automatically at steady state (§2.4)—whose node voltages *are* the initial conditions:

|            |                                                |
|------------|------------------------------------------------|
| e,s,0,18   | <i>the DC source, still connected</i>          |
| r8,s,a,8   | <i>...behind its <math>8 \Omega</math></i>     |
| c1,a,0,1/6 | <i>first capacitor (open at steady state)</i>  |
| r12,a,0,12 | <i>the resistive ladder...</i>                 |
| r18,a,b,18 |                                                |
| r6,b,0,6   |                                                |
| c2,b,0,1/3 | <i>second capacitor (open at steady state)</i> |

returns  $v_a = 9 \text{ V}$  and  $v_b = \frac{9}{4} \text{ V}$ . The  $t > 0$  circuit drops the source branch, connects the waveform source, and carries those two numbers as the capacitors’ initial-condition fields—the fifth field that has existed since the five-column matrices of 1999:

|                                                  |                                                                    |
|--------------------------------------------------|--------------------------------------------------------------------|
| c1,a,0,1/6,9                                     | <i>first capacitor, <math>v(0) = 9 \text{ V}</math></i>            |
| r12,a,0,12                                       | <i>the ladder, source branch dropped...</i>                        |
| r18,a,b,18                                       |                                                                    |
| r6,b,0,6                                         |                                                                    |
| c2,b,0,1/3,9/4                                   | <i>second capacitor, <math>v(0) = \frac{9}{4} \text{ V}</math></i> |
| js,0,b,10*e <sup>^</sup> (-t)*sin(2*t+30*pi/180) | <i>the waveform source, now connected</i>                          |

A TR analysis (§3.3) recognizes the source value as a waveform, transforms it in, solves in  $s$  with the initial conditions inside the capacitor stamps, and inverts every answer. The result,

for  $t \geq 0$ :

$$v_1(t) = \frac{193 + 160\sqrt{3}}{34} e^{-t/2} + \frac{153 - 170\sqrt{3} + 20 \cos(2t + \frac{\pi}{6}) - 80 \sin(2t + \frac{\pi}{6})}{34} e^{-t}$$

$$\approx 13.83 e^{-t/2} - 4.16 e^{-t} + e^{-t} [0.588 \cos(2t + 30^\circ) - 2.353 \sin(2t + 30^\circ)] \text{ V}.$$

The structure is legible at a glance: the circuit's two natural modes ( $e^{-t/2}$  and  $e^{-t}$ ) and the forced response riding on the source's own  $e^{-t}$  envelope. And the answer carries its own check: evaluating at  $t = 0$  gives exactly 9 V—the capacitor voltage is continuous across the switching instant, as physics demands.

Now the comparison this exemplar exists for. The thesis solved this same problem on a TI-89 in 2001—same DC run for the initial conditions (9 V and  $\frac{9}{4}$  V, printed), then an FD run with the (Impala) mode waking for the waveform source—and printed the expanded answer [2, §12.3]:

$$v_1(t) \approx -0.667 e^{-t} \cos 2t - 2.33 e^{-t} \sin 2t + 13.8 e^{-t/2} - 4.16 e^{-t}.$$

Expanding version 9's exact answer above into the same form gives  $-0.667 e^{-t} \cos 2t - 2.332 e^{-t} \sin 2t + 13.83 e^{-t/2} - 4.16 e^{-t}$ : every coefficient agrees with the 2001 printout to the digits the calculator displayed. Twenty-five years apart, one machine a 12 MHz handheld and the other a computer algebra system, the two implementations of the logic this monograph documents return the same circuit answer—which is the strongest single piece of evidence we can offer that the logic, and not any one program, is the durable thing.

Done by hand, this problem is an afternoon of Laplace algebra with two initial conditions threaded through every step; with Symbulator it is two descriptions and two calls, and the student's attention stays where Prof. Boulet presumably wanted it—on the switching, the steady states, and the modes.

## 5.4 Three-phase: wye–delta with line impedances

Three-phase problems are the traditional graveyard of sign errors: three sources at  $120^\circ$  intervals, line impedances, a delta load, and a page of phasor mesh algebra between the question and the answer. This one is Alexander & Sadiku's Example 12.11 [10], a balanced wye–delta system *with* line impedances—the variant textbooks save for last. In Symbulator the sources are typed the way the book prints them, angle notation and all:

```
ea1,na1,0,100           the wye source, phase a...
eb1,nb1,0,(100∠−120°)   ...phase b...
ec1,nc1,0,(100∠120°)    ...phase c
raa,na1,na2,1           the three line impedances...
rbb,nb1,nb2,1
rcc,nc1,nc2,1
rac,na2,nc2,100+24*pi*j  the delta load...
rcb,nc2,nb2,100+24*pi*j
rba,nb2,na2,100+24*pi*j
```

One AC call returns every line current and node phasor; in polar display, the *a*-line current is  $i_{raa} = 2.350 \angle -36.20^\circ$  A and the load's line voltage  $v_{na2} - v_{nb2} = 169.94 \angle 30.81^\circ$  V. The exemplary part is what is *absent*: no wye–delta transformation, no single-phase equivalent, no bookkeeping of which angle belongs to which mesh—the standard manoeuvres exist to make hand solution tractable, and a solver that simply writes all the equations does not need them. The student can instead spend the attention where the three-phase insight lives: on the  $\sqrt{3}$  and  $30^\circ$  relationships visible between the answers.

## 5.5 The ideal transformer, symbolically

From the version 9 teaching corpus (after Alexander & Sadiku's Fig. 13.33): a source  $v_{s2}$  behind an impedance  $z_2$  on the secondary of an ideal  $1:n$  transformer, viewed from the primary terminals. Everything in the description is a name:

|                         |                                          |
|-------------------------|------------------------------------------|
| <code>t,2,3,1,n</code>  | <i>the ideal 1:n transformer</i>         |
| <code>r2,3,4,z2</code>  | <i>the secondary impedance, symbolic</i> |
| <code>e2,4,0,vs2</code> | <i>the secondary source, symbolic</i>    |

Asking for the Thévenin equivalent seen at the primary port—`TH` at  $(2,0)$ , Chapter 4—returns

$$v_{th} = \frac{v_{s2}}{n}, \quad z_{eq} = \frac{z_2}{n^2}.$$

That is, the tool *derives* the referred-source and referred-impedance identities every machines course teaches: a voltage refers by the turns ratio, an impedance by its square. When the answer to a symbolic question is the textbook's own theorem, the simulator has done something a numeric tool cannot: it has produced knowledge in the same form the student is supposed to carry away.

## 5.6 Coupled coils with initial conditions

Composed for this monograph: two identical coils,  $L_1 = L_2 = 2$  H, coupled with  $M = 1$  H, each closed on its own  $3\ \Omega$  resistor; no sources anywhere. The primary carries 1 A at  $t = 0$ ; the secondary is at rest. What happens?

|                         |                                              |
|-------------------------|----------------------------------------------|
| <code>l1,1,0,2,1</code> | <i>primary coil, <math>i(0) = 1</math> A</i> |
| <code>r1,1,0,3</code>   | <i>...closed on its resistor</i>             |
| <code>l2,2,0,2</code>   | <i>secondary coil, at rest</i>               |
| <code>r2,2,0,3</code>   | <i>...closed on its resistor</i>             |
| <code>m,11,12,1</code>  | <i>the coupling, <math>M = 1</math> H</i>    |

The initial condition rides in the inductor's optional field, exactly where it has lived since the five-column matrices of 1999, and the FD stamps carry it as the  $i(0)/s$  terms of §3.2—including inside the *mutual* terms, which is the part that is miserable to do by hand. One TR call:

$$i_{L1}(t) = \frac{e^{-t} + e^{-3t}}{2}, \quad i_{L2}(t) = \frac{e^{-t} - e^{-3t}}{2}, \quad v_2(t) = \frac{3(e^{-3t} - e^{-t})}{2}.$$

The physics is legible in the algebra: the circuit's two modes are the *common* mode (currents aiding, effective inductance  $L + M = 3$  H,  $\tau = 1$  s) and the *differential* mode ( $L - M = 1$  H,  $\tau = \frac{1}{3}$  s), and the answers are exactly their sum and difference. The secondary current starts at zero, rises as the fast mode dies, peaks, and decays with the slow mode—energy handed across the coupling and returned to heat, the whole story in two exponentials.

## 5.7 Two op-amps, all symbols

From the teaching corpus (after Bobrow's Example 3.3 [11]): a two-op-amp summing cascade in which every conductance is a symbol,  $g_1, g_2, g_3, g_4, g$ , and the source is  $v_s$ :

|              |                                                     |
|--------------|-----------------------------------------------------|
| e,1,0,vs     | <i>the source, symbolic</i>                         |
| r12,1,2,1/g1 | <i>conductance <math>g_1</math>, into stage one</i> |
| r14,1,4,1/g2 | <i>conductance <math>g_2</math>, into stage two</i> |
| r23,2,3,1/g  | <i>stage one's feedback, <math>g</math></i>         |
| r34,3,4,1/g  | <i>interstage coupling, <math>g</math></i>          |
| r2o,2,o,1/g4 | <i>feedback from the output, <math>g_4</math></i>   |
| r4o,4,o,1/g3 | <i>feedback from the output, <math>g_3</math></i>   |
| o1,0,2,3     | <i>first nullor</i>                                 |
| o2,0,4,o     | <i>second nullor</i>                                |

Two nullors, seven symbolic elements, one DC call:

$$v_o = v_s \frac{g_1 - g_2}{g_3 - g_4}.$$

This is the answer a designer wants, in the form a designer wants it: the output is a *difference of conductances over a difference of conductances*, so the structure of the circuit is read directly off the expression—set  $g_1 = g_2$  and the output nulls; let  $g_3 \rightarrow g_4$  and the gain diverges; trim any one conductance and the sensitivity is a one-line partial derivative. A numeric simulation of the same circuit would have produced one number per run and none of this shape. Symbolic answers on op-amp networks are, in the author's experience, the single most persuasive demonstration of what Symbulator is *for*.

# Chapter 6

## The users, over the decades

Software for students lives or dies by whether students reach for it, and Symbulator’s story is best told in its users’ own words, collected over more than twenty-five years [1, 4, 8].

### 6.1 The early years

From the first year, the reports had a pattern: relief at the mathematics lifted away, and surprise that a calculator could do it. Chris Riegel, then president of the IEEE student chapter at the University of Kentucky, wrote: “For EE211 and EE221, Symbulator actually works better than SPICE because you get the results symbolically. Once you use Symbulator you’ll see what everybody has been talking about. If you’re not using Symbulator, you’re working too hard.” Jay Myers, of the University of Texas at Arlington, reviewed it at length: “Once a circuit is solved, it has all the information the user would want: node voltages, element currents, element power usage, voltage drops... It solves symbolically, unlike almost any other circuit simulator (even PC based). This program is, hands down, the best program for circuit analysis that I have seen.” [1]

The Spanish-language site gathered the same story from Latin America: thirteen users in one group at the University of Central Florida (“those who use all three programs always say yours is the easiest and fastest”); thanks from Chile “for making life easier for all of us who study Electrical Engineering”; congratulations from Mexico on a manual “perfectly well explained” [4]. Users reported from over twenty countries on five continents, and the program spread the way student tools spread: one classmate at a time.

The thesis’s acknowledgments page is a snapshot of that community at its two-year mark, and reads like a map [2, Acks.]: Tim Hutcheson, Joe Riel, Charles Ware, Doug Burkett, Jake Adams, Al Charpentier, and Nevin McChesney in the United States; Lars Frederiksen in Denmark; Arne Harstad in Norway; Erwin Baert in Belgium; Reinhard Willinski in Germany; Kamil Malinski in Austria; Daniele Martini in Italy; Rozgonyi Szabolcs in Hungary; Michael Rans in the United Kingdom; Alex Astashyn in Russia; Pepe Iborra in Spain; Ivan Oro Yu in Panama—thanked by name “for stimulating me with their suggestions and comments, to develop, improve and test Symbulator during these years.” Several appear elsewhere in this monograph as more than names: Frederiksen wrote the Laplace engine the calculator versions ran on, Baert the first **par** and **absang**, Riel the simulator that inspired the whole project, Burkett and Hutcheson the English translation of the thesis itself. Two of the thesis’s 89 problems came from users (McChesney’s and Adams’s), recorded in its provenance appendix like any textbook source [2, App. D]: the community was, from the start, inside the documentation and not merely around it.

### 6.2 The long run

A quarter century later, the collected praise reads like a single voice across generations [8]: “the masterpiece of TI-89 programming” (Alex Astashyn, New York); “Symbulator is and always will be the best program for TI-89/92/V200” (Charles Ware, Ohio); “it gives you every-

thing you need... DC, AC, transient analysis, Thévenin, Bode plots... everything!” (Nikolaos Trichakis, Thessaloniki); “I wish I had it while going through school” (Joe Riel—the same Joe Riel whose Syrup had pointed the way); “Symbulator is worth the cost of a TI-89” (Pier Giorgio Raponi, Pisa); “it certainly is helping me to focus on my studying of EE instead of trying to remember all the math” (Jim Wachtel, Missouri). Several of these correspondents—Frederiksen, Hutcheson, Burkett, Riel among them—became collaborators, translators, and friends; the full roll is kept in the Symbulator documentation’s credits [8].

## 6.3 Closing reflections

The users’ words converge on the claim this monograph opened with, and the internal logic presented here explains *why* the tool earns them.

*Symbols first.* Because the engine is symbolic end to end, a student can solve the circuit *as drawn in the textbook*—with  $R_1$ ,  $L$ , and  $v_S$  as letters—and watch the structure of the answer: which elements appear, which cancel, how a pole arises. A numeric simulator answers “what”; a symbolic one also answers “why,” and the difference is precisely the part a course wants students to internalize.

*The input is the concept.* The description language is a one-to-one transcription of a schematic: an element, its terminals, its value. Dependent sources are ordinary values that name an answer; the levels mirror how a hand solution actually proceeds (voltages, then currents, then powers). The equations generated are the ones a student would write. Nothing in the interface is an artifact to learn and later unlearn.

*Failure is didactic.* The solver’s refusals teach: it names the loop of sources that contradicts itself, the node whose current sources cannot balance, the floating sub-circuit with no reference. The same care appears at the input (an ambiguous 1k, quoted back exactly as typed) and in the tools (an unbounded short-circuit current reported as the real answer that it is). Error paths designed this way convert the classic frustrations of simulation into circuit-theory lessons.

*Verification at scale.* Because answers are exact, the whole of Symbulator’s teaching corpus is machine-checkable against the documentation’s printed answers—and is checked, example by example. A student can trust that the tool and the book agree, and an instructor can extend the corpus knowing that regressions announce themselves. This, too, is the thesis’s stance carried forward: the simulator’s first duty is *verification*—of the student’s own hand work, and even of the textbook’s printed answers, “some of which are wrong” [2, Recs.].

*Knowing when to decline.* The thesis’s most disarming appendix asks when it is better *not* to use Symbulator [2, App. C], and answers with a worked problem that a single `solve` command dispatches faster than any simulation could. The skill it names—“a kind of intuition that tells you whether, given the nature of a problem, it will be more complicated to solve by simulation than by hand”—is one a tool’s own author rarely thinks to teach, and its presence in the foundational document says much about the project’s temperament: the tool was built to serve the analysis, never to replace the judgment.

There is also a lesson in the project’s longevity. The same internal logic has now run on 1990s calculator hardware and on a modern computer algebra system, with the level architecture surviving as a *conceptual* account even where it is no longer an operational necessity. Designs that mirror the structure of the subject matter—here, of circuit analysis as actually taught—age well; the hardware around them is what changes. And the most recent chapter, a two-decade-old port completed in weeks by its author working with an AI assistant, suggests that long-lived educational software can be carried to each new platform at a fraction of its original cost, so long as its logic was worth preserving in the first place.

*The purpose of circuit analysis is insight, not mathematics*—and the purpose of Symbulator, across every platform it has lived on, has been to keep the two in their right order.

# Appendix A

## The historical record

Two documents anchor the history told in Chapter 1, and both are reproduced here in facsimile from the author’s papers.

The first (Figure A.1) is the official record of the thesis defense—*Informe de Sustentación del Trabajo de Graduación*, Universidad Tecnológica de Panamá, Facultad de Ingeniería Eléctrica—for the graduation work titled *Symbulator: un simulador de circuitos lineales para calculadoras*, defended 9 July 2001. The examining tribunal’s three members—Medardo Logreira, Eliane de Cabrera, and Marcela de Vásquez—each recorded the grade *A*, *100*, for a final average of 100, and wrote the general observations quoted in translation in §1.4: an excellent work, an innovative contribution to numerical and symbolic circuit simulation, recognized at the world level, a pride of the Faculty and of the University, recommended for promotion within the University community.

The second (Figure A.2) is the certificate of the IEEE award mentioned in that verdict’s “recognition at the world level”: First Place in the 2000 Region 9 Student Paper Competition, for the paper *Symbulator: A Symbolic Simulator of Circuits for Calculators*—the 1999 competition paper [1], submitted under the pseudonym *Thévenin* and unmasked at the awarding. The certificate, dated October 2000 and signed by the IEEE’s Secretary and President, spells *Symbolic* with the Spanish *í*: a small orthographic fingerprint of a Panamanian student’s English-language debut, preserved here as issued.

UNIVERSIDAD TECNOLÓGICA DE PANAMÁ  
SECRETARÍA GENERAL

FORM N°  
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INF. N° \_\_\_\_\_

**INFORME DE SUSTENTACION DEL TRABAJO DE GRADUACION**

|                                                                                                                                                                                                                                                                                                                                                                                                            |             |                |         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------|---------|
| NOMBRE DEL GRADUANDO <b>ROBERTO PEREZ-FRANCO</b>                                                                                                                                                                                                                                                                                                                                                           |             |                |         |
| NUMERO DE CEDULA <b>6-700-1101</b>                                                                                                                                                                                                                                                                                                                                                                         |             |                |         |
| FACULTAD <b>INGENIERIA ELECTRICA</b>                                                                                                                                                                                                                                                                                                                                                                       |             |                |         |
| CARRERA <b>LICENCIATURA EN INGENIERIA ELECTROMECHANICA</b>                                                                                                                                                                                                                                                                                                                                                 |             |                |         |
| TITULO QUE ASPIRA <b>LICENCIATURA EN INGENIERIA ELECTROMECHANICA</b>                                                                                                                                                                                                                                                                                                                                       |             |                |         |
| TITULO DEL TRABAJO DE GRADUACION <b>SYMBULATOR</b>                                                                                                                                                                                                                                                                                                                                                         |             |                |         |
| <b>un simulador de circuitos lineales para calculadoras .</b>                                                                                                                                                                                                                                                                                                                                              |             |                |         |
| ESTE TRABAJO FUE SUSTENTADO EN EL <b>Primer</b> SEMESTRE DEL AÑO <b>2001</b>                                                                                                                                                                                                                                                                                                                               |             |                |         |
| CODIGO DE ASIGNATURA <b>4391</b> CODIGO DE HORARIO _____                                                                                                                                                                                                                                                                                                                                                   |             |                |         |
| TRIBUNAL                                                                                                                                                                                                                                                                                                                                                                                                   | CALIFICADOR | FIRMA          | PUNTAJE |
| NOMBRE                                                                                                                                                                                                                                                                                                                                                                                                     |             |                |         |
| 1. Medardo Logreño                                                                                                                                                                                                                                                                                                                                                                                         |             | <i>[Firma]</i> | A 100   |
| 2. Elise de Cabrera                                                                                                                                                                                                                                                                                                                                                                                        |             | <i>[Firma]</i> | A 100   |
| 3. Marcela de Vasquez                                                                                                                                                                                                                                                                                                                                                                                      |             | <i>[Firma]</i> | A 100   |
| PROMEDIO FINAL                                                                                                                                                                                                                                                                                                                                                                                             |             |                | A 100   |
| OBSERVACIONES GENERALES DEL TRIBUNAL CALIFICADOR                                                                                                                                                                                                                                                                                                                                                           |             |                |         |
| Es un excelente trabajo que provee un aporte innovador en el campo de la simulación numérica y simbólica de los circuitos eléctricos y que ha recibido reconocimiento a nivel mundial. Este trabajo es un orgullo para la Facultad de Ingeniería Eléctrica y para la Universidad Tecnológica de Panamá. Los miembros del jurado recomiendan que se promocióne este trabajo entre la comunidad Tecnológica. |             |                |         |
| FECHA DE SUSTENTACION <b>9/7/2001</b> Vo. Bo. DEL DECANO <i>[Firma]</i>                                                                                                                                                                                                                                                                                                                                    |             |                |         |

NOTA: EL INFORME DEBE SER REMITIDO A LA SECRETARIA GENERAL A MAS TARDAR OCHO (8) DIAS DESPUES DE LA FECHA DE SUSTENTACION.

Figure A.1: The jury record of the thesis defense, 9 July 2001: three grades of 100/100, and the tribunal's citation.

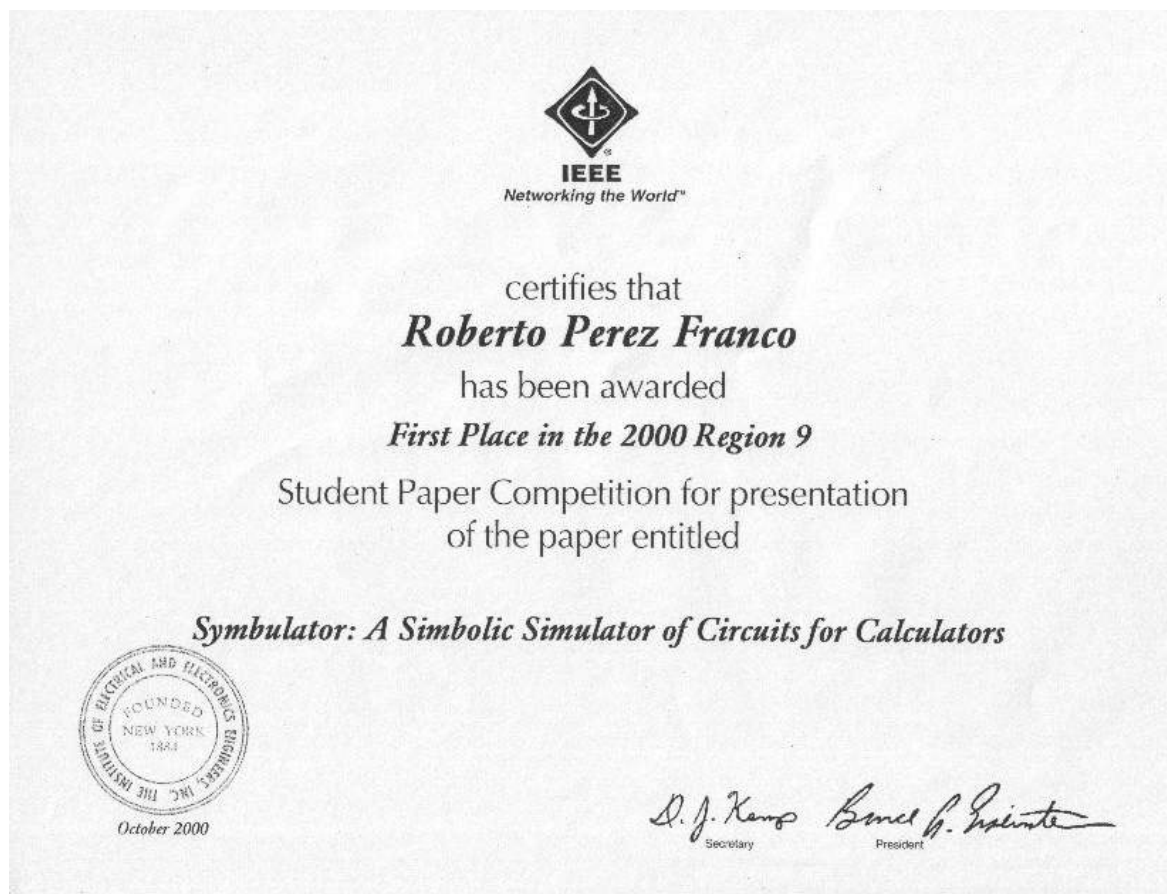


Figure A.2: First Place, IEEE Region 9 Student Paper Competition, 2000.

## Appendix B

# The exemplar circuits, drawn

The schematics below are not illustrations prepared for this monograph: each one was *drawn* by *Symbulator 9's schematic engine* from the same circuit description that Chapter 5 solves, exactly as the app draws a circuit beside its answers. The conventions are the drawer's own: wires never cross an element body, genuine crossings are drawn as semicircular hops, junction dots sit on node corners, values read as the user typed them, and a dependent source carries its controlling expression as its value.

## Acknowledgments

Symbulator 9 is built on Python and SymPy, and Symbulator's author thanks their creators and maintainers. Versions 7 and 8's Laplace transforms are due to Lars Frederiksen, adapted for the TI-Nspire by Philippe Fortin. Over the decades, many users improved Symbulator through suggestions and corrections; they are acknowledged individually in the Symbulator documentation. The teaching corpus draws its problems from widely used circuit analysis textbooks, credited in full in the documentation.

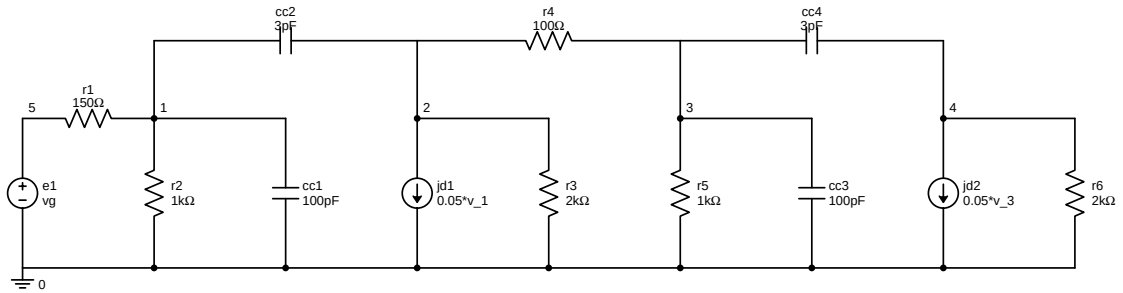


Figure B.1: The two-stage amplifier of 1999 (§5.1); thesis Problem 87.

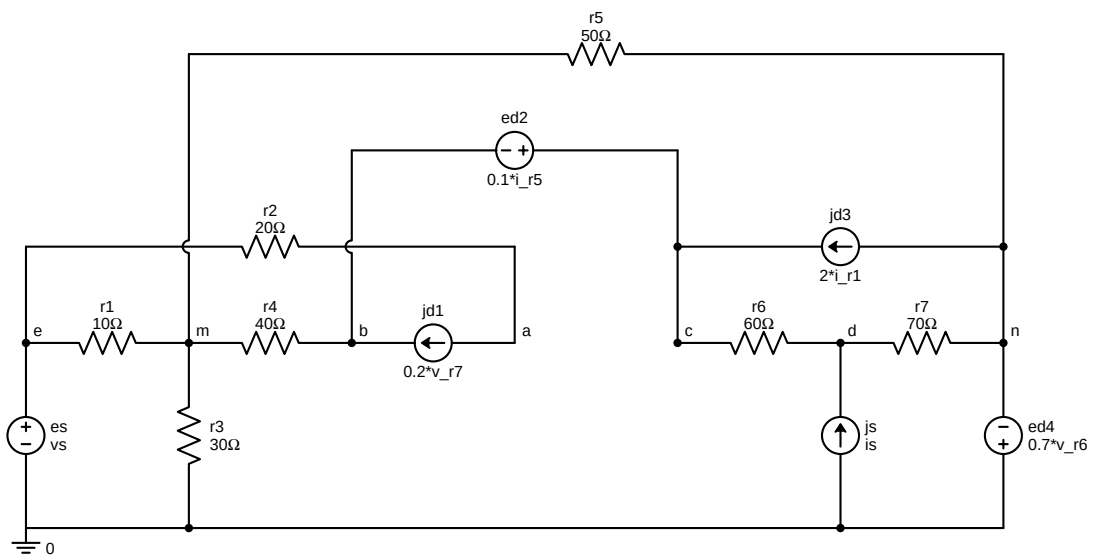


Figure B.2: The 2013 showcase (§5.2): one dependent source of each species, and both independent sources symbolic.

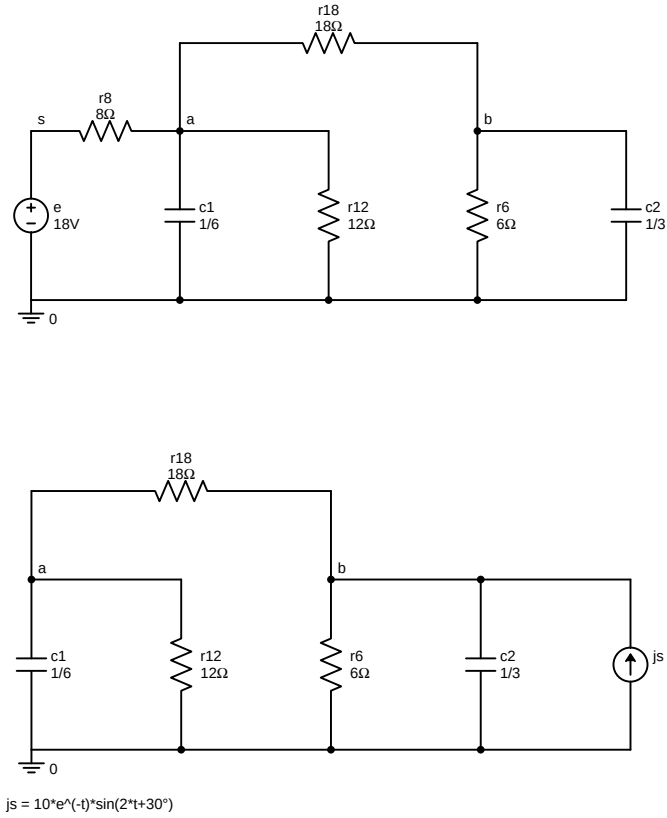


Figure B.3: Prof. Boulet’s switching transient (§5.3); thesis Problem 88. Above, the  $t < 0$  circuit whose DC solution supplies the initial conditions; below, the  $t > 0$  circuit carrying them in the capacitors’ fifth fields, with the waveform source connected.

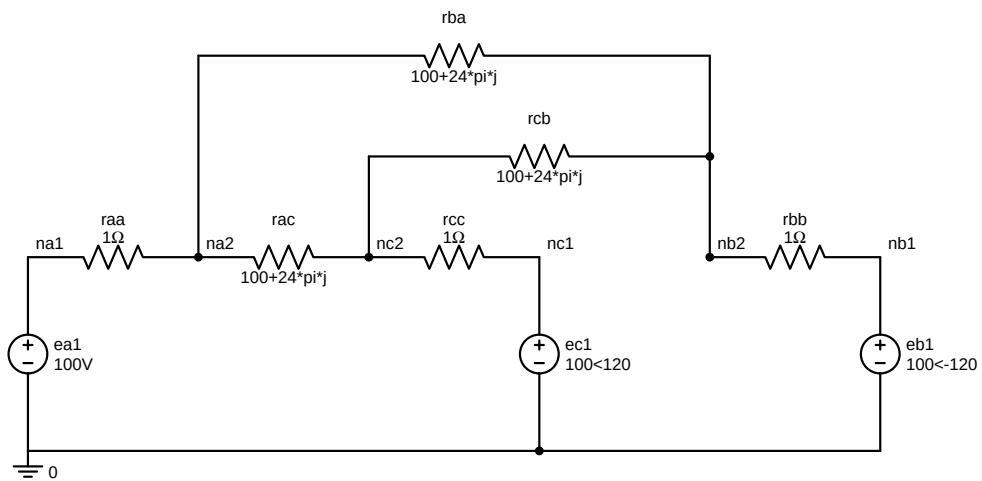


Figure B.4: The balanced wye–delta system with line impedances (§5.4), its sources typed in angle notation.

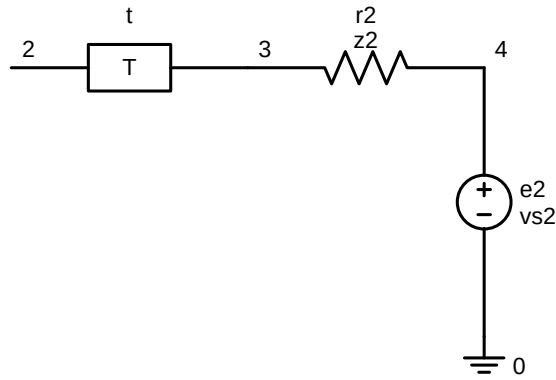


Figure B.5: The ideal transformer viewed from its primary (§5.5), every value a symbol.

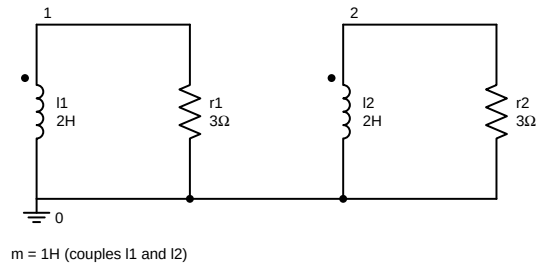


Figure B.6: The coupled coils with an initial current (§5.6).

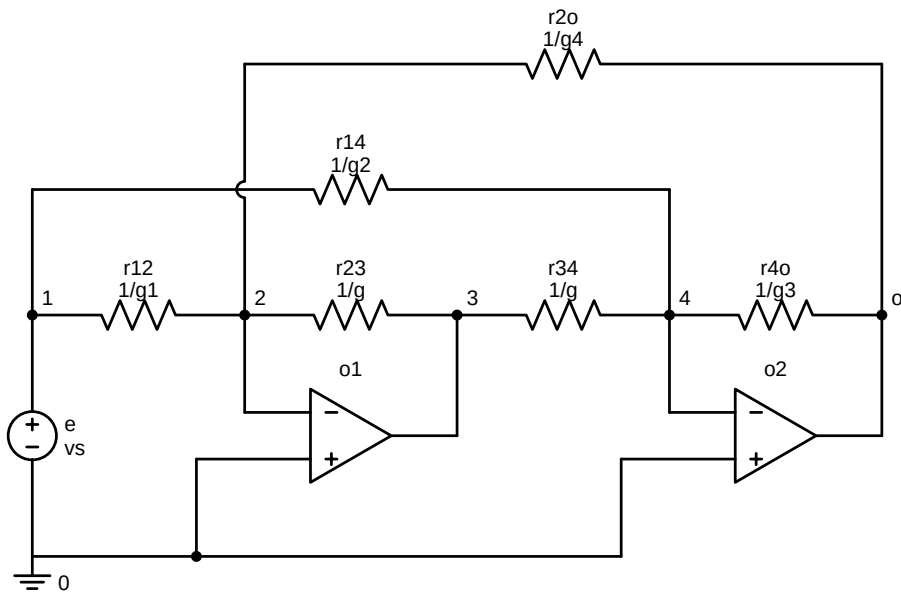


Figure B.7: The all-symbolic two-op-amp cascade (§5.7).

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- [3] R. Pérez-Franco, *The Symbulator Book: A Linear Circuit Simulator for Calculators*, the completed English edition of the 2001 thesis [2]. Preface, notice, and chapters 1–4 translated by D. Burkett and T. Hutcheson (ca. 2001); chapters 5–12, back matter, and appendices translated by Claude (2026) in the same voice. Published alongside this monograph. [Online]. Available: <https://learn.symbulator.com/book.pdf>
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